Anytime-Valid Sequential Testing of Group Synergy: Interventional and Predictive Formulations, with an Application to Financial Markets

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Abstract

Does a *group* of variables act on an outcome jointly, beyond the sum of its members' separate contributions, and can this be decided *as data arrive*, stopping when the evidence is conclusive? Existing synergy measures (sufficient-cause interaction indices, joint-effect estimators, information-theoretic decompositions) are fixed-sample. We introduce ANTE, an anytime-valid sequential test of a super-additive synergy null built on testing by betting: a bounded interaction score is accumulated into a nonnegative capital process (an e-process), and Ville's inequality gives time-uniform Type-I error control under optional stopping. In the *interventional* part (known randomised design) the null is sequential, $\boldsymbol{\theta_t \leq 0}$ for all $\boldsymbol{t}$, so stationarity is not required. In the *predictive* part (ANTE-SG, for time series) the null compares two learned forecasters, an additive and a joint one, through a truncated-loss score whose bound is known before each observation; this makes the validity proof exact and replaces a clipping argument that we show to be invalid in general. We prove validity for the forecaster-comparison null, power one and a high-probability detection-time bound for an oracle bet, a $\boldsymbol{k}$-way extension, and bounds on the size of synergy that can be excluded, and we state sufficient conditions under which a rejection can be read as an $\boldsymbol{A \times B}$ interaction. Simulations show that the procedure holds its level under continuous monitoring, including heavy-tailed, correlated and nonstationary nulls, but that its power on real financial noise is low: an injected lagged interaction carrying **40%** of the variance of a daily return series is detected in fewer than half of real triples. Type-I control for population-level nulls is checked by simulation and by placebo calibration on real data, not proved. In a fully specified search of **27,027** hypotheses over financial panels we find no statistically significant super-additive predictive group effect in daily and weekly returns and volatility, where

the evidence matches a placebo null; for stress indicators and hourly crypto data the test rejects a true placebo null in **10–40%** of hypotheses, and we do not interpret rejections there as synergy.

Keywords: anytime-valid inference, e-processes, testing by betting, synergy, group synergy, predictive synergy, Granger causality

MSC Classification: 62L10 , 62M10 , 62G10 , 91G70

1 Introduction

Some effects require a coalition. Two drugs that are inert alone can be lethal together; two genes silent in isolation can jointly switch on a phenotype; two market shocks, each absorbed on its own, can compound into a crisis. The common structure is *synergy*: the joint action of a set of variables exceeds the sum of their individual actions. Distinguishing "you need *both* A and B" from "A helps and B helps, separately" is a recurring scientific question, and a subtle one, because it is a statement about an *interaction*, not about any single variable.

The question has been formalised at least three times, in largely separate literatures. In epidemiology, Rothman's sufficient-component-cause model (Rothman 1976) defines synergy as two factors being component causes of the same causal "pie", measured by the additive-scale interaction contrast and its relative-excess-risk index (Hosmer and Lemeshow 1992; VanderWeele and Knol 2014), with a complete counterfactual axiomatisation for arbitrarily many binary causes due to VanderWeele and Richardson (2012) and, more recently, a multi-factor generalized synergy index (La Torre and D'Urso 2026). In statistics, a parallel programme estimates the effect of a *joint* intervention $\mathrm{do}(\{A, B, \dots\})$ by fusing experiments (Roquero Gimenez and Rothenhäusler 2023) or identifying the joint law of potential outcomes (Wu and Mao 2025). In information theory, Partial Information Decomposition (Williams and Beer 2010) and its dynamical descendants SURD (Martínez-Sánchez et al. 2024) and PEID (Yang et al. 2026) split the information that sources carry about a target into redundant, unique, and *synergistic* parts.

These methods share one limitation that this paper addresses: they are **fixed-sample**. Each returns a point estimate, and at best a confidence interval, at a sample size chosen *before* looking at the data. None supports the mode of inference that streaming data demand: watch the evidence accumulate and stop when it is decisive, without paying a multiplicity penalty for having looked. A practitioner who applies an unadjusted fixed-sample synergy test repeatedly ("peeking") generally loses its nominal Type-I error; we show below that for the repeated Wald procedure the error rises from a nominal 5% to about 50%.

The machinery for continuous monitoring now exists. Testing by betting (Shafer 2021), e-processes and safe anytime-valid inference (Ramdas et al. 2023; Grünwald et al. 2024), and confidence sequences (Howard et al. 2021) provide tests whose validity holds *simultaneously at all sample sizes* and under *any* data-dependent stopping rule.

A sceptic bets against the null; if the null is true the sceptic's capital is a nonnegative supermartingale and cannot grow, so large capital is evidence against the null, quantified by Ville's inequality (Ville 1939). This programme has produced anytime-valid tests for means (Waudby-Smith and Ramdas 2024), off-policy and average treatment effects (Waudby-Smith et al. 2024), comparisons of sequential forecasters (Choe and Ramdas 2024), conditional independence (Grünwald et al. 2024) and kernelised independence (Podkopaev et al. 2023). To the best of our knowledge, it has not been applied to an interaction or synergy contrast.

What is, and is not, claimed.

We keep four notions apart throughout. *Interaction* is a property of a contrast ($\mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}$ in the interventional part, a difference of predictive gains in the predictive part). *Synergy* is a positive (super-additive) interaction. *Causal* synergy needs an interventional or counterfactual definition and a design that identifies it; the interventional part supplies both under a known randomised design. *Predictive* synergy, tested in the time-series part, is a statement about out-of-sample forecasting: it is Granger-type predictability and is not, by itself, evidence of a structural interaction. We therefore speak of "predictive group synergy" for the time-series procedure and reserve "causal" for the interventional one.

Contributions.

The paper makes the following contributions.

1. **A sequential test of synergy with an exact validity proof, in two formulations.** (i) Interventional ANTE: under a known, possibly adaptive, randomised design we build a bounded interaction score that is centred at the conditional contrast θ_t; the one-sided null $\theta_t \leq 0$ for all t gives a nonnegative supermartingale (Theorem 2) and, via Ville's inequality, time-uniform Type-I control (Corollary 3). The null is sequential, so nonstationarity of the contrast is allowed. (ii) Predictive ANTE-SG: the null compares two learned forecasters, and a truncated-loss score normalised by a bound that is known before each observation gives an exactly valid e-process (Theorem 8). This replaces a clipping construction that we show to fail in general (Example 1). Part A is a specialisation to a contrast vector of the anytime-valid average-effect e-processes of Waudby-Smith et al. (2024); its novelty is the target (an interaction) and the sequential null, not the betting machinery.
2. **Power theory, stated for what it covers.** For an oracle fixed bet we prove power one (Theorem 5) and a non-asymptotic, high-probability detection-time bound (Proposition 6); the bound extends to a mixture over a fixed grid of bets (Proposition 7). For the implemented plug-in bet we prove validity only, and evaluate power empirically.
3. **Groups, exclusion and multiplicity.** A k-way extension contrasts an additive with an all-interaction forecaster (Corollary 9); an *exclusion* e-process against persistent synergy (Proposition 11) and a time-uniform upper bound on the running average of the conditional synergy (Proposition 12) quantify what can be ruled out; e-Bonferroni and e-BH control family-wise error and FDR across many hypotheses.

4. **A precise isolation analysis.** We show when the add-versus-joint contrast isolates an $A \times B$ interaction from joint dependence and own-nonlinearity (Proposition 10), prove that the four-model contrast Syn_4 does not (it can be negative or positive under purely additive mechanisms with correlated sources), and give counterexamples showing that additivity alone is not enough.
5. **An empirical programme with an explicit search space and honest calibration.** Simulations cover heavy-tailed, correlated, nonstationary and nonlinear alternatives, with Monte Carlo uncertainty throughout. A formally specified financial search over daily, weekly and hourly panels, three target types and four lag structures (27,027 hypotheses) finds no family-wise detection in daily and weekly returns and volatility, and the real-data evidence there coincides with a placebo null; the placebo also shows that the test is *not* calibrated for binary stress indicators or hourly crypto series, so rejections there are not interpreted. We report the power of the test on real noise, which is low, and we show that physical turbulence data, a natural positive control, fail a placebo check.

Section 2 positions the work; Section 3 fixes notation and the two nulls; Section 4 develops the interventional test and Section 5 the predictive test, including a table of what is proved and what is only evaluated empirically (Table 2). Section 6 reports the experiments, and Section 7 discusses limitations. Proofs are in the appendices.

2 Related work and positioning

Group causality, the notion that a set of causes acts on an effect jointly beyond the sum of its members, has been studied in four research traditions. Three formalise *what synergy is* (epidemiological sufficient-cause interaction, statistical joint-effect estimation, and information-theoretic decomposition); the fourth supplies the inferential machinery we import (game-theoretic sequential testing).

2.1 Sufficient-cause and additive-scale interaction

Rothman's *sufficient-component-cause* model (Rothman 1976) defines synergy as two factors being component causes of a common sufficient cause. For two binary causes with counterfactual cell means $\mu_{ab} = \mathbb{E}[C(a, b)]$, this is captured on the additive (risk-difference) scale by the interaction contrast $\theta = \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}$; the *relative excess risk due to interaction* (RERI) was placed on a firm inferential footing by Hosmer and Lemeshow (1992) and systematised by VanderWeele and Knol (2014). VanderWeele and Richardson (2012) give empirical and counterfactual conditions for sufficient-cause and *singular* interaction among an arbitrary number of dichotomous exposures; this is the batch null our interventional test most directly generalises. Mechanistic interaction has its own measures, such as the peril-ratio index PRISM (Lee 2013) and its gene–gene extension (Lin and Lee 2018), and La Torre and D'Urso (2026) generalise Rothman's synergy index to many factors. All of these are fixed-sample.

2.2 Joint effects, causal aggregation, and multi-cause identification

A second tradition asks what the effect of intervening on several causes at once is. Roquero Gimenez and Rothenhäusler (2023) develop *causal aggregation* for the effect of a joint intervention from experiments that each manipulate few variables; Wu and Mao (2025) study nonparametric identification of the joint distribution of potential outcomes; and deep structural-equation approaches (Balgi et al. 2024) estimate interventional and counterfactual distributions. These are effect-estimation methods with a fixed design; none provides a sequentially valid test of a synergy contrast.

2.3 Information-theoretic synergy and higher-order interactions

Interaction information (McGill 1954) was the first attempt to isolate the information that three or more variables share beyond their pairwise relations. *Partial Information Decomposition* (PID; Williams and Beer 2010) splits the information that two sources carry about a target into redundant, unique and synergistic atoms; Lyu et al. (2026) show that the lattice becomes inconsistent for four or more variables. The O-information (Rosas et al. 2019) indicates whether a system is net-synergistic or net-redundant. For dynamical systems, SURD (Martínez-Sánchez et al. 2024) decomposes transfer-entropy-style causality into synergistic, unique and redundant components, and PEID (Yang et al. 2026) provides an interventional analogue of PID in which synergy is $EI(A, B \to C) - \sum_i EI(T^{(i)} \to C)$. These quantities answer a different question from the one tested here: they measure how much information is jointly carried, under their own definition of synergy, and they are *estimators* rather than tests, with no null distribution, calibrated error rate or stopping rule. We compare against them in Section 6.3 as complementary tools, not as competitors on a common estimand.

2.4 Synergy and causality in time series and finance

Directed predictability is formalised by Granger causality (Granger 1969; Geweke 1982) and, information-theoretically, by transfer entropy (Schreiber 2000); the two coincide for Gaussian variables (Barnett et al. 2009). Stramaglia et al. (2014) show that pairwise Granger analysis misses synergy and that conditioning reveals it. In finance, Scagliarini et al. (2020) compute PID synergy over triplets of market indices, and the connectedness literature measures system-wide spillovers through variance decompositions (Diebold and Yılmaz 2014) and through pairwise Granger-causality networks (Billio et al. 2012). These works measure synergy or connectedness in sample, at a fixed sample size, without an out-of-sample, error-controlled test. Closer in spirit to our empirical design, Choi (2026) studies time-varying lag structures of Bitcoin's Kimchi premium with rolling-window Granger causality and transfer entropy, using a stationary bootstrap with pre-specified lag grids, multiple-testing corrections and winsorisation sensitivity because the series are strongly leptokurtic; we adopt the same disciplines (pre-specified lag structures, winsorisation and truncation sensitivity, window-level reporting) for a group-level test.

2.5 Testing by betting and anytime-valid inference

The inferential machinery we import is game-theoretic statistics (Ramdas et al. 2023): Ville's inequality (Ville 1939), testing by betting (Shafer 2021), safe testing (Grünwald et al. 2024), the hedged-capital betting strategies for bounded means of Waudby-Smith and Ramdas (2024), time-uniform confidence sequences (Howard et al. 2021) and universal inference (Wasserman et al. 2020). Continuous-monitoring validity distinguishes these from classical tests, whose guarantees degrade under repeated looking (Johari et al. 2022); the classical remedy is a group-sequential design with alpha spending (Pocock 1977; O'Brien and Fleming 1979; Lan and DeMets 1983), which we use as a comparator. E-values can be merged under arbitrary dependence (Vovk and Wang 2021) and support FDR control via e-BH (Wang and Ramdas 2022). Two lines are close to our predictive part: sequential comparison of forecasters (Choe and Ramdas 2024), whose null we adopt (Section 5), and anytime-valid conditional-independence and kernelised independence tests (Grünwald et al. 2024; Podkopaev et al. 2023), which could supply a causal or nonlinear route that we leave to future work. The distributional Granger test of Jha (2026) is a sequential test of a *single* source.

2.6 The gap, and the claim

We distinguish five properties: (P1) anytime-valid inference for an interaction contrast; (P2) anytime-valid *causal* inference; (P3) sequential inference for effects of *groups*; (P4) predictive synergy; (P5) information-theoretic synergy. Table 1 records which of them each strand addresses. A focused search (terms: *anytime-valid interaction test*, *e-process causal interaction*, *sequential test synergy*, *confidence sequence interaction effect*, *test martingale synergistic*) did not return a sequential, anytime-valid test whose null is that the joint effect of a set of variables equals the sum of its parts. A negative search result cannot be exhaustive, and the literature on e-values and online testing is growing quickly. We therefore make the following restricted claim: *to the best of our knowledge, ANTE is the first framework we are aware of that combines (P1) and (P3), with an interventional (P2, under a known design) and a predictive (P4) instantiation.* We do not claim to be the first anytime-valid test of any synergy-like quantity, and we do not address (P5) beyond comparing with it.

3 Setup: two formulations of the synergy null

We develop the null on two scales that share the idea *joint equals sum of parts* but differ in what is being estimated and what is assumed. They are kept separate in the theory (Sections 4 and 5) because they support different interpretations.

3.1 Interventional synergy

Observations arrive as a stream with unit t carrying (X_t, A_t, B_t, C_t), where $C_t \in [0,1]$ is a bounded outcome, $A_t, B_t \in \{0,1\}$ are binary causes, and X_t are optional covariates. Let $\mathcal{F}_t = \sigma\big((X_s, A_s, B_s, C_s)_{s \le t}\big)$ and $\mathcal{H}_t = \mathcal{F}_{t-1} \vee \sigma(X_t)$. Let $C_t(a,b)$ be the potential outcome under $\mathrm{do}(A = a, B = b)$.

Table 1 Properties addressed by each strand. (P1) anytime-valid inference for an interaction contrast; (P2) anytime-valid causal inference; (P3) sequential inference for groups; (P4) predictive synergy; (P5) information-theoretic synergy. Entries describe the cited works as we read them.

Strand	P1	P2	P3	P4	P5
Sufficient-cause / RERI / synergy index (§2.1)	×	×	×	×	×
Joint effects / causal aggregation (§2.2)	×	×	×	×	×
PID / SURD / PEID / O-information (§2.3)	×	×	×	×	✓
Time-series and finance synergy (§2.4)	×	×	×	✓ (in sample)	✓ (in sample)
Anytime-valid effects, forecaster comparison (§2.5)	×	✓ (single effect)	×	×	×
ANTE (this paper)	✓	✓ (known design)	✓	✓	×

Assumption 1 (Known sequential design) For every t: (i) *consistency without carry-over*, $C_t = C_t(A_t, B_t)$, so potential outcomes at time t depend only on the causes assigned at time t (no interference between periods); (ii) *known sequential randomisation*, $\mathbb{P}(A_t = a, B_t = b \mid \mathcal{H}_t, \{C_t(\cdot,\cdot)\}) = \pi_t(a,b)$ with π_t known and $\mathcal{H}_t$-measurable (adaptive designs are allowed); (iii) *positivity*, $\pi_t(a,b) \geq \pi_{\min} > 0$.

Definition 1 (Conditional additive interaction contrast)

$$\theta_t(X_t) = \sum_{a,b} c(a,b)\,\mathbb{E}\big[C_t(a,b) \mid \mathcal{H}_t\big], \qquad \theta_t = \mathbb{E}\big[\theta_t(X_t) \mid \mathcal{F}_{t-1}\big],$$

with $c(a,b) = (-1)^{(1-a)+(1-b)}$, so that $c = +1$ on the cells $(1,1), (0,0)$ and -1 on $(1,0), (0,1)$. If the cell means do not depend on t, $\theta_t \equiv \mu_{11} - \mu_{10} - \mu_{01} + \mu_{00}$.

The *sequential* nulls are

$$H_0^{\leq}: \ \theta_t \leq 0 \text{ for all } t \quad \text{(one-sided: no super-additive synergy)},$$
$$H_0^{=}: \ \theta_t = 0 \text{ for all } t \quad \text{(two-sided)}.$$

They allow the contrast to vary over time: a fixed-parameter null $\theta = 0$ is the special case $\theta_t \equiv \theta$. The scientific question of super-additive synergy is inherently one-sided, and $H_0^{\leq}$ is the null we use in the applications; the two-sided null is needed only to detect antagonism. For example, an XOR mechanism has $\theta = -2$ on this additive scale (the cell means of the concordant cells are zero and those of the discordant cells one), so it is a *negative* additive interaction; information-theoretic decompositions call XOR "pure synergy" because neither source alone is informative, which is a different notion.

3.2 Predictive group synergy for time series

Observational streams admit no intervention, so we use an out-of-sample predictive definition in the spirit of Granger (Granger 1969). Let Y be a target, A, B candidate sources, Z an optional conditioning set (for example a market factor), and p the lag order. For $S \subseteq \{A, B\}$ let $L(S)$ be the expected one-step loss of the best (linear least-squares) predictor of Y_t from the $\{Y, Z\}$-past augmented by the lagged histories of S, and $\Delta(S) = L(\varnothing) - L(S)$ its predictive gain. Two contrasts are natural:

$$\mathrm{Syn}_4(A, B \to Y) = \Delta(\{A, B\}) - \Delta(\{A\}) - \Delta(\{B\}), \tag{1}$$

$$\mathrm{Syn}_{\mathrm{aj}}(A, B \to Y) = L(\mathrm{add}) - L(\mathrm{joint}), \tag{2}$$

where "add" is the best predictor from the base set plus *both* sources' own terms (linear and squared lags) and "joint" additionally contains the cross-products $A_{t-\ell}B_{t-\ell}$. $\mathrm{Syn}_{\mathrm{aj}}$ is the *add-versus-joint* contrast. Syn_4 is the more obvious interaction of predictive gains; we show in Section 5.3 that it is contaminated by redundancy between A and B and should not be used to isolate interaction. In practice the test is run on the *learned* online forecasters, whose null we state in Section 5.1.

Both Syn_4 and $\mathrm{Syn}_{\mathrm{aj}}$ are predictive quantities. A positive value says that the joint past of (A, B) forecasts Y better than an additive combination; it does not by itself identify a structural interaction. In particular, for Gaussian variables Granger causality equals transfer entropy (Barnett et al. 2009), so Syn can be read as an interaction of conditional transfer entropies, but that reading is predictive, not causal.

4 Part A: interventional ANTE

4.1 A bounded score, centred at the conditional contrast

Under Assumption 1 define the inverse-probability-weighted interaction score

$$\psi_t \;=\; c(A_t, B_t)\,\frac{C_t}{\pi_t(A_t, B_t)}. \tag{3}$$

Lemma 1 (Conditional unbiasedness and boundedness) *Under Assumption 1,* $\mathbb{E}[\psi_t \mid \mathcal{H}_t] = \theta_t(X_t)$ *and hence* $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \theta_t$. *Moreover* $|\psi_t| \le c_\psi := 1/\pi_{\min}$*; for a balanced* 2×2 *design* $\psi_t \in [-4, 4]$.

The bound, not any variance or tail assumption, is what makes the betting argument exact. An augmented (AIPW) score is discussed in Section 4.4.

4.2 The betting e-process and its validity

Fix a level $\alpha \in (0,1)$. Start with unit capital and bet a *predictable* fraction λ_t, that is, an $\mathcal{F}_{t-1}$-measurable function of the past:

$$E_t \;=\; \prod_{s=1}^{t}\bigl(1+\lambda_s\psi_s\bigr), \qquad \lambda_s \in \Bigl[0, \frac{\gamma}{c_\psi}\Bigr] \;\text{(one-sided)}, \;\; \gamma < 1. \tag{4}$$

Theorem 2 (Validity, nonstationary null) *Under Assumption 1:*

(a) (one-sided) if $\lambda_s \in [0, \gamma/c_\psi]$, then under $H_0^{\leq}$ the process (E_t) is a nonnegative supermartingale with $E_0 = 1$;

(b) (two-sided) let E_t^+ and E_t^- be built from predictable $\lambda_s^+ \in [0, \gamma/c_\psi]$ and $\lambda_s^- \in [-\gamma/c_\psi, 0]$. Under $H_0^{=}$ the average $\frac{1}{2}(E_t^+ + E_t^-)$ is a nonnegative martingale with mean 1.

Corollary 3 (Anytime-valid Type-I control) *In either case, $\mathbb{P}\bigl(\exists\, t \geq 1 : \; E_t \geq 1/\alpha\bigr) \leq \alpha$. The guarantee holds uniformly over all stopping times: one may monitor after every observation, stop early, or continue indefinitely. E_τ is an e-value at every stopping time τ, and $1/\max_{s\leq t} E_s \wedge 1$ is an anytime-valid p-value.*

Remark 1 (Scope of the two-sided process) The two-sided average is valid under $H_0^{=}$ but *not* under $H_0^{\leq}$: if $\theta_t < 0$ the leg with $\lambda \leq 0$ has positive drift. We therefore use the one-sided process for the question "is there super-additive synergy?".

Remark 2 (Why peeking a fixed-sample test fails) A fixed-n test controls $\mathbb{P}(\text{reject at } n)$ for one pre-committed n. Consulting an unadjusted fixed-sample test after every observation reports $\mathbb{P}(\exists t : \text{reject at } t)$, which in general exceeds the nominal level and grows with the number of looks. Corollary 3 is the statement such a procedure lacks. We quantify the inflation for a repeated Wald test in Section 6.1; that procedure is included to illustrate the optional-stopping failure, not as a competing anytime-valid method.

The choice of bet.

Any predictable λ_t is valid; a *good* one grows capital fast under the alternative. We use a truncated plug-in Kelly fraction from the running moments of ψ,

$$\lambda_t = \min\Bigl\{\max\Bigl(\frac{\hat\mu_{t-1}}{\hat v_{t-1}}, 0\Bigr), \frac{\gamma}{c_\psi}\Bigr\}, \qquad \hat\mu_{t-1} = \tfrac{1}{t-1}\sum_{s<t}\psi_s, \quad \hat v_{t-1} = \tfrac{1}{t-1}\sum_{s<t}\psi_s^2, \tag{5}$$

the aGRAPA-type strategy of Waudby-Smith and Ramdas (2024) specialised to the interaction score, with $\lambda_t = 0$ during a warm-up of t_0 steps and whenever $\hat v_{t-1} = 0$. The warm-up, the cap γ and the moment estimators are part of the definition of the e-process, not of the algorithm only, and are fixed before data are seen.

4.3 Power and detection time (oracle bet)

Validity without power is vacuous, so we state what can be proved about power, and for which bet. The results below hold for an *oracle fixed bet* λ (or a fixed grid); validity of arbitrary predictable bets, power of the oracle bet, and the behaviour of the implemented plug-in bet are three different things, and we prove only the first two.

Assumption 2 (Persistent alternative) $|\psi_t| \le c$ and $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] \ge \delta > 0$ for all t.

Lemma 4 *For $\gamma \in [0,1)$ and $x \ge -\gamma$, $\log(1+x) \ge x - C_\gamma x^2$ with $C_\gamma = \frac{-\log(1-\gamma)-\gamma}{\gamma^2} \le \frac{1}{2(1-\gamma)}$ (and $C_0 = \frac{1}{2}$).*

Theorem 5 (Power one for the oracle bet) *Under Assumption 2, fix $\lambda \in (0, \gamma/c]$ and put $\bar\gamma = \lambda c$. Then almost surely*

$$\liminf_{t\to\infty} \frac{1}{t} \log E_t \;\ge\; g_0(\lambda) := \lambda\delta - C_{\bar\gamma}\lambda^2 c^2 \;\ge\; \lambda\delta - \frac{\lambda^2 c^2}{2(1-\gamma)}.$$

If $g_0(\lambda) > 0$ then $E_t \to \infty$ a.s., so the anytime-valid test rejects with probability one. The cap-based Kelly choice $\lambda^\star = \delta(1-\gamma)/c^2$, admissible when $\delta(1-\gamma) \le \gamma c$ (always if $\gamma \ge \frac{1}{2}$ and $\delta \le c$), gives $g_0 \ge \delta^2(1-\gamma)/(2c^2)$.

Proposition 6 (Non-asymptotic detection-time bound) *Under Assumption 2, for a fixed λ with $g_0 = g_0(\lambda) > 0$ and $\ell_\lambda = \log\frac{1+\lambda c}{1-\lambda c}$, the rejection time $\tau_\alpha = \inf\{t : E_t \ge 1/\alpha\}$ satisfies, for every $t > \log(1/\alpha)/g_0$,*

$$\mathbb{P}(\tau_\alpha > t) \;\le\; \exp\Big(-\frac{2\,(t\,g_0 - \log(1/\alpha))^2}{t\,\ell_\lambda^2}\Big).$$

Consequently, with probability at least $1-\beta$, $\tau_\alpha \le t_\beta = \max\big\{2\log(1/\alpha)/g_0,\; 2\ell_\lambda^2\log(1/\beta)/g_0^2\big\}$. At the cap-based Kelly bet this is of order $\log(1/\alpha)/((1-\gamma)\delta^2)$ for the first term and $\log(1/\beta)/\delta^2$ for the second.

This is a high-probability bound, not an almost-sure bound and not a bound on $\mathbb{E}\tau_\alpha$: detection time scales as δ^{-2} in the effect and only logarithmically in $1/\alpha$.

Proposition 7 (Mixture over a fixed grid of bets) *Let $0 < \lambda_1 < \cdots < \lambda_K \le \gamma/c$ be fixed before the data are seen and $E_t^{\mathrm{mix}} = K^{-1}\sum_j \prod_{s\le t}(1+\lambda_j\psi_s)$. Then E^{mix} is a nonnegative supermartingale under $H_0^{\le}$, and $\log E_t^{\mathrm{mix}} \ge \max_j \log E_t(\lambda_j) - \log K$. Hence Theorem 5 and Proposition 6 hold for the best grid point with $\log(1/\alpha)$ replaced by $\log(K/\alpha)$. On the geometric grid $\lambda_j = (\gamma/c)2^{-(j-1)}$, $j = 1,\dots,K$, with K large enough that $\lambda_K \le \lambda^\star$, the rate loss relative to the quadratic lower bound $\lambda\delta - C_\gamma\lambda^2c^2$ at its maximiser is at most a factor $4/3$.*

For the implemented plug-in bet (5) we prove validity only; its power and detection time are evaluated in simulation (Section 6.1). Regret guarantees for related betting strategies exist (Waudby-Smith and Ramdas 2024), but we do not claim them for the specific truncated plug-in used here.

4.4 Augmented (AIPW) scores

Let $\phi_t = \sum_{a,b} c(a,b)\big[\frac{\mathbb{1}\{A_t=a,B_t=b\}}{\pi_t(a,b)}\big(C_t - \hat{\mu}_{ab,t}(X_t)\big) + \hat{\mu}_{ab,t}(X_t)\big]$ with outcome models $\hat{\mu}_{ab,t} \in [0,1]$ fitted on past data only (so $\mathcal{F}_{t-1}$-measurable). Under a *known* design $\mathbb{E}[\phi_t \mid \mathcal{H}_t] = \theta_t(X_t)$ and $|\phi_t| \le 1/\pi_{\min} + 2$, so Theorem 2 holds for ϕ_t with c_ψ replaced by $1/\pi_{\min} + 2$, whatever the quality of $\hat{\mu}$ (Appendix A). With an *estimated* propensity score the centring error is second-order but non-zero, so finite-sample validity is not guaranteed; we present that case as an extension and leave its finite-sample theory to future work.

5 Part B: predictive ANTE-SG for time series

5.1 The null is a comparison of two forecasters

At each t two forecasts of Y_t are made from information available before Y_t is observed: p_t^{add} from the additive model and p_t^{joint} from the joint model. Both are produced by online forgetting recursive-least-squares (RLS) learners (Ljung 1999) updated only after each outcome. Let $\ell(\cdot)$ be a loss and let the per-step score be $s_t = \ell(Y_t - p_t^{\text{add}}) - \ell(Y_t - p_t^{\text{joint}})$; $s_t > 0$ means the joint forecaster did better. The null is

$$H_0^{\text{preq}}: \qquad \mathbb{E}\big[s_t \mid \mathcal{F}_{t-1}\big] \le 0 \quad \text{for all } t, \tag{6}$$

a prequential (forecaster-comparison) null in the sense of Dawid (1984) and Choe and Ramdas (2024). It is a statement about the *realised, learned* forecasters, and requires no stationarity and no correct specification of the data-generating process.

Remark 3 (What H_0^{preq} is not) H_0^{preq} is not the population statement $\text{Syn}_{\text{aj}} = 0$. If $\mu_t = \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}]$, $e_a = p_t^{\text{add}} - \mu_t$ and $d = p_t^{\text{joint}} - p_t^{\text{add}}$, then for the squared loss $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] = -2de_a - d^2$, which is positive if and only if $de_a < -d^2/2$, that is, when the joint refit moves the forecast in the direction opposite to the additive model's current estimation error by less than $2|e_a|$. Averaged over estimation noise it is negative (the out-of-sample penalty of the larger nested model, Clark and West 2007), but that is an average, not a conditional statement, and forgetting learners move slowly, so favourable stretches can persist. A rejection therefore means: *the joint forecaster had a persistently positive conditional predictive advantage over the additive forecaster*. Non-rejection and exclusion statements (Section 5.5) are statements about these forecasters; they bear on population synergy only under the conditions of Section 5.3, which are checked in simulation but cannot be verified in data.

Filtration and predictability.

Let $\mathcal{F}_t$ be the natural filtration of the raw observations (of the whole panel, when many hypotheses are scanned). Every ingredient used to form E_t is $\mathcal{F}_{t-1}$-measurable at the time the bet is placed: (i) the standardisation of each series uses only running moments of past values; (ii) the features are fixed functions of lagged standardised values; (iii) the RLS state is a fixed function of past features and outcomes, and the code predicts before updating, so $p_t^k = \langle w_{t-1}^k, \phi_t^k \rangle$; (iv) the normaliser b_t below is a function of

the predictions; (v) the bet λ_t is a function of $u_1, \dots, u_{t-1}$. All tuning constants (forgetting factor, ridge, lag order, truncation level, bet cap, warm-up, feature maps and any random-feature frequencies) are fixed before any data are seen; selecting any of them using the series under test voids the guarantee. If features contain contemporaneous values (A_t, B_t, Z_t), the null and b_t are conditioned on $\mathcal{G}_t = \mathcal{F}_{t-1} \vee \sigma(A_t, B_t, Z_t)$ and validity follows by the tower property; such a test concerns contemporaneous interaction and not Granger-type prediction.

5.2 A bounded score with a predictable bound

A natural construction scales the raw squared-error score by a predictable standard deviation and clips it, arguing that clipping and positive scaling preserve the sign of the conditional mean. That argument is false in general, and we do not use it.

Example 1 (Clipping can reverse the sign of the conditional mean) Let s take the value -20 with probability 0.1 and $+1.5$ with probability 0.9, with $\hat{\sigma} = 1$ and clip level $b = 5$. Then $\mathbb{E}s = -0.65 \le 0$ but $\mathbb{E}\,\mathrm{clip}(s/\hat{\sigma}, -b, b) = 0.1(-5) + 0.9(1.5) = +0.85 > 0$. A bettor that wagers a fixed fraction on i.i.d. draws grows its capital at a positive exponential rate, so the clipped test rejects a true raw null with probability one. A data-dependent scale does not repair this once the tail is heavier (Appendix A); we reproduce the failure in simulation in Section 6.1.

Clipping is valid for the null stated *on the clipped score*, and for the raw null under conditional symmetry (Appendix A); neither is what a user of the test wants. We replace clipping by a *truncated loss* and a *predictable bound*. Let $\ell_c(e) = \min(e^2, c^2)$ for a truncation level $c > 0$ (in units of the predictably standardised target). Because ℓ_c is $2c$-Lipschitz, for any two predictable forecasts and every outcome

$$\left|\ell_c(Y_t - p_t^1) - \ell_c(Y_t - p_t^2)\right| \;\le\; 2c\,|p_t^1 - p_t^2|. \tag{7}$$

The bound does not use nesting and is known *before* Y_t is observed. Define

$$\begin{aligned} s_t = \ell_c(Y_t - p_t^{\mathrm{add}}) - \ell_c(Y_t - p_t^{\mathrm{joint}}), \qquad b_t = 2c\,|p_t^{\mathrm{add}} - p_t^{\mathrm{joint}}|, \\ u_t = \frac{s_t}{b_t}\,\mathbb{1}\{b_t > \varepsilon\} \in [-1, 1], \end{aligned} \tag{8}$$

with a fixed threshold $\varepsilon \ge 0$ (if $b_t = 0$ the two forecasts coincide and $s_t = 0$). Dividing by the predictable positive b_t preserves the sign of the conditional mean *exactly*: $\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[s_t \mid \mathcal{F}_{t-1}]\mathbb{1}\{b_t > \varepsilon\}/b_t$. No clipping and no distributional assumption are needed.

Theorem 8 (Exact validity of ANTE-SG) *Let $\lambda_t \in [0, \gamma]$ be predictable, $\gamma < 1$, and $E_t = \prod_{s \le t}(1 + \lambda_s u_s)$. Under H_0^{preq}, (E_t) is a nonnegative supermartingale with $E_0 = 1$. Hence $\mathbb{P}(\exists t : E_t \ge 1/\alpha) \le \alpha$, and E_τ is an e-value for every stopping time τ. The same holds for the capped process $\min\{E_{t-1}(1 + \lambda_t u_t), C\}$ for any constant $C \ge 1$ (the implementation uses $C = 10^{15}$).*

Corollary 9 (k-way groups) *For a group $\mathcal{G} = \{A_1, \ldots, A_k\}$ let p^{add} come from the feature map Φ^{add} (each member's own linear and squared lags) and p^{joint} from Φ^{joint} which adds all same-lag products of order $2, \ldots, k$. Then Theorem 8 holds verbatim under the group null $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \le 0$, with $b_t = 2c|p_t^{\text{add}} - p_t^{\text{joint}}|$.*

The same construction applies to the four-model score $s_t = \ell_c^A + \ell_c^B - \ell_c^{\text{joint}} - \ell_c^{\text{base}}$ with $b_t = 2c(|p^A - p^{\text{joint}}| + |p^B - p^{\text{base}}|)$; we keep it only as a diagnostic.

5.3 What the add-versus-joint contrast isolates

Theorem 8 is valid for H_0^{preq} whatever the data. What *rejection means* depends on the population structure, and it is easy to over-state it. Work in the least-squares projection framework: second-order stationarity, finite fourth moments, untruncated squared loss, population-optimal linear predictors on the feature blocks of Section 3.2.

Proposition 10 (Isolation, corrected) *(a)* $\text{Syn}_{\text{aj}} \ge 0$, *and* $\text{Syn}_{\text{aj}} = 0$ *if and only if every cross-product $A_{t-\ell}B_{t-\ell}$ is orthogonal to the residual of the additive projection.*

(b) $\text{Syn}_4 = \text{Syn}_{\text{aj}} - R$, *where the redundancy term R vanishes when the two sources' feature blocks are orthogonal after conditioning on the base, but otherwise can have either sign. For jointly Gaussian (A, B) with correlation ρ and $Y = A \pm B + \varepsilon$ one has $\text{Syn}_{\text{aj}} = 0$ while $\text{Syn}_4 = -2\rho(1+\rho)$ for the plus sign and $+2\rho(1-\rho)$ for the minus sign.*

(c) Under an additive mechanism $Y_t = m(W_t) + f(\mathcal{A}_t) + h(\mathcal{B}_t) + \varepsilon_t$ (with $\mathbb{E}[\varepsilon_t \mid W_t, \mathcal{A}_t, \mathcal{B}_t] = 0$ and finite fourth moments), $\text{Syn}_{\text{aj}} = 0$ in each of the following cases: (S1) $m(W)$ lies in the span of the base features and f and h lie in the own-feature spans (linear plus quadratic lags), with arbitrary dependence between A and B; (S2) the base, the A-past and the B-past are mutually independent (f, h arbitrary; then $R = 0$ too); (S3) one lag, (A, B) jointly Gaussian and independent of the base (f, h arbitrary, cubic terms included).

(d) Without such conditions $\text{Syn}_{\text{aj}} > 0$ is possible under a mechanism that involves a single source. For example, with $A, V \sim N(0,1)$ and ε independent, $B = A^2 + \tau V$ and $Y = A^3 + \varepsilon$ gives $\text{Syn}_{\text{aj}} = 36/(6+\tau^2)$; a fully Gaussian two-lag example ($p = 2$, A_t, V_t i.i.d. $N(0,1)$), $B_t = A_{t-1} + vV_t$ and $Y_t = A_{t-1}A_{t-2} + \varepsilon_t$, gives $\text{Syn}_{\text{aj}} = 1/(1+v^2)$.

Hence an e-process rejection of H_0^{preq} can be read as evidence of an irreducible $A \times B$ interaction only under conditions such as (c); when sources are dependent and the mechanism may be nonlinear outside the feature span, it is evidence that the joint forecaster beats the additive one. We therefore do not claim that the test "provably isolates" interaction from joint dependence; what holds is Proposition 10 and the simulations of Section 6.2.

5.4 Estimation noise and negative evidence

The joint forecaster estimates more coefficients than the additive one. With an effective window of $n_{\text{eff}} = (1+\rho)/(1-\rho)$ observations (about 400 for a forgetting factor $\rho = 0.995$) this estimation noise does not vanish, and it gives the score a negative average drift even when the population contrast is zero or small and positive. Validity is unaffected (a more negative drift only slackens the supermartingale inequality), but it follows that a small final capital $E_t < 1$ carries no evidential weight *for* the null: e-values below one are not evidence of absence. We report a sensitivity analysis over ρ and the Clark–West adjusted score $s_t + (p_t^{\text{add}} - p_t^{\text{joint}})^2$ (Clark and West 2007) as a diagnostic of the noise floor. The adjusted score is *not* guaranteed to be centred, so it is not used for inference.

5.5 Exclusion: bounding the size of synergy

A rejection cannot be used the other way round, so absence statements need a separate instrument. Two are provided: an e-process against a persistent alternative, and a time-uniform upper bound on the running average of the conditional synergy.

Units.

The score is a difference of *truncated* squared losses of a predictably standardised target, so its unit is a per-step gain in truncated squared error per unit target variance. Only about 2% of the observations of the series used below lie beyond three predictable standard deviations, but they carry 34–63% of the sum of squares (Section 6.5), so a gain of δ in these units is not comparable to a gain in untruncated out-of-sample R^2 by a comparable factor; we call it "loss units".

Persistent alternative.

For $\delta > 0$ let $H_\delta : \mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \geq \delta$ for all monitored t. With $w_t(\delta) = (\delta - s_t)/(b_t + \delta) \in (-1, 1]$ and predictable $\lambda_t \in [0, \gamma]$, put $K_t(\delta) = \prod_{s \leq t}(1 + \lambda_s w_s(\delta))$.

Proposition 11 (Exclusion e-process) *Under H_δ, $K(\delta)$ is a nonnegative supermartingale, so $\mathbb{P}(\exists t : K_t(\delta) \geq 1/\alpha) \leq \alpha$. For a finite grid G of values of δ with per-δ bets, rejecting every δ with $\max_t K_t(\delta) \geq |G|/\alpha$ makes the probability of rejecting any true H_δ in the grid at most α.*

Rejecting H_δ excludes *persistent* synergy of at least δ at every monitored time; it does not exclude synergy confined to a regime or an average synergy of that size. Because $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] \leq b_t$, H_δ forces $b_t \geq \delta$, so the process wins mechanically on steps where the two learners nearly agree. On real data, with learners that agree to within $0.0005/2c$ at some steps, this made the persistent exclusion uninformative (nearly every hypothesis excluded the smallest δ on the grid), and we therefore do not use it for the finance conclusions.

Running-average synergy.

Let $m_s = \mathbb{E}[s_s \mid \mathcal{F}_{s-1}]$, $S_t = \sum_{s=t_1}^{t} s_s$ and $V_t = \sum_{s=t_1}^{t} b_s^2$ over the monitored steps $s \geq t_1$ (t_1 fixed in advance, after the warm-up).

Proposition 12 (Time-uniform bound on average synergy) *Let $\lambda_1, \ldots, \lambda_K > 0$ be a grid fixed before the data are seen. With probability at least $1 - \alpha$, simultaneously for all $t \geq t_1$ and all j,*

$$\frac{1}{t - t_1 + 1} \sum_{s=t_1}^{t} m_s \;\leq\; \frac{S_t}{t - t_1 + 1} + \frac{\lambda_j V_t/2 + \log(K/\alpha)/\lambda_j}{t - t_1 + 1}$$

(and the symmetric lower bound). The inequality holds without any null assumption; it bounds the average of the conditional means of the score, in loss units, for the realised forecasters. For m hypotheses replace α by α/m.

This is the bound reported in Section 6.5. It is valid whatever the data are, but it concerns the conditional mean of the score under the learned forecasters and shares with the other results the reading of Remark 3.

5.6 Many hypotheses

For m hypotheses with e-processes $E^{(i)}$, each a supermartingale under its null, rejecting H_i when $\sup_t E_t^{(i)} \geq m/\alpha$ (e-Bonferroni) controls the family-wise error rate at level α under arbitrary dependence. e-BH (Wang and Ramdas 2022) controls the FDR at level $\alpha|I_0|/m$ when fed e-values $E_\tau^{(i)}$ at a stopping time τ (never the running maximum, which is not an e-value); if τ depends on all streams, each null is conditioned on the whole-panel past. The price is a threshold of m/α: $\log_{10}(m/\alpha) = 4.1$ for $m = 660$ and 4.9 for $m = 4400$, against 1.3 at $m = 1$, so detection takes roughly 3–4 times longer, or the detectable δ is 1.8–2 times larger. Zero family-wise detections are therefore weak evidence of absence on their own, and absence statements use the time-uniform bound on average synergy of Proposition 12, with the same multiplicity correction.

5.7 Algorithm and a summary of guarantees

6 Experiments

The programme answers five questions in turn. Does the theory hold, including under nulls that stress it (§6.1)? What does the add-versus-joint score isolate when sources are correlated or the mechanism is nonlinear (§6.2)? How does the test compare with batch estimators and with a classical group-sequential design (§6.3)? Does it find interactions that are present in real data (§6.4)? And what does a fully specified search of financial data show, and how much can it exclude (§6.5)? Every number below can be regenerated from the scripts in `experiments/revision/`.

Defaults.

Level $\alpha = 0.05$; the test rejects when the running maximum of the e-process reaches $1/\alpha = 20$ (equivalent to the first-crossing rule). ANTE-SG uses lag order $p = 1$ unless

Algorithm 1 Interventional ANTE (one-sided)

Require: stream (X_t, A_t, B_t, C_t); known design π_t; level α; cap $\gamma < 1$; warm-up t_0

$E \leftarrow 1$; running $\hat{\mu}, \hat{v} \leftarrow 0$
for $t = 1, 2, \ldots$ **do**
  $\psi_t \leftarrow c(A_t, B_t)\, C_t / \pi_t(A_t, B_t)$ ▷ Eq. (3)
  $\lambda_t \leftarrow \min\{\max(\hat{\mu}/\hat{v}, 0), \gamma/c_\psi\}$ **if** $t > t_0$ **else** 0 ▷ predictable
  $E \leftarrow E(1 + \lambda_t\, \psi_t)$
  if $E \geq 1/\alpha$ **then return reject** $H_0^{\leq}$ at time t
  end if
  update $\hat{\mu}, \hat{v}$ with ψ_t
end for

Algorithm 2 ANTE-SG (predictive group synergy, streaming)

Require: panel Y, A, B, conditioner Z; lag p; level α; forgetting ρ; truncation c; cap γ; warm-up t_0

standardise each series predictably; init two forgetting-RLS learners (add, joint); $E \leftarrow 1$
for $t = p + 1, \ldots, T$ **do**
  form p_t^{add}, p_t^{joint} from information up to $t - 1$ ▷ predict before updating
  $b_t \leftarrow 2c|p_t^{\text{add}} - p_t^{\text{joint}}|$; observe Y_t; $s_t \leftarrow \ell_c(Y_t - p_t^{\text{add}}) - \ell_c(Y_t - p_t^{\text{joint}})$; $u_t \leftarrow s_t / b_t$ if $b_t > \varepsilon$ else 0
  $\lambda_t \leftarrow \min\{\max(\hat{\mu}_{t-1}/\hat{v}_{t-1}, 0), \gamma\}$ **if** $t > t_0$ **else** 0
  $E \leftarrow E\,(1 + \lambda_t u_t)$
  if $E \geq 1/\alpha$ **then return joint forecaster advantage detected** at t
  end if
  update both learners and the running moments of u with $(\text{features}_t, Y_t)$
end for

stated, a forgetting factor $\rho = 0.995$ (effective window about 400 steps), ridge 10^{-2}, truncation level $c = 3$ in units of the predictably standardised target, bet cap $\gamma = 0.9$, warm-up $t_0 = 50$, and feature winsorisation at ± 5 for real data. Real series are standardised with running moments of the past (30-observation warm-up). The conditioner is the market factor (SPY, or its volatility/stress transform for those targets) and the same transform is applied to every series of a panel. Rates are reported with Wilson 95% intervals (Wilson 1927). In figures and tables "original test" denotes the clipped-score procedure (Example 1), "four-model" the contrast of Eq. (1), and "ANTE-SG" the add-versus-joint test of Theorem 8. Capital is capped at 10^{15}, so $\log_{10} E = 15$ means "at or above the cap".

Table 2 Guarantees. "Proved" refers to the results in this paper under the stated assumptions; "empirical" means evaluated only in simulation or on data.

Property	Status	Conditions
Interventional validity (one-sided, nonstationary)	Proved (Thm 2)	known sequential design, positivity; $\theta_t \leq 0\ \forall t$
Interventional validity (two-sided)	Proved	as above with $\theta_t = 0\ \forall t$
AIPW score, known design	Proved	past-fitted outcome models
AIPW, estimated propensity	Not proved	extension, asymptotic only
ANTE-SG validity (truncated loss, bound b_t)	Proved (Thm 8)	predictable forecasts, tuning fixed in advance; null H_0^{preq}
Clipped-score validity	Fails in general (Ex. 1)	valid only for the clipped null or under conditional symmetry
Power one, detection time	Proved (Thm 5, Prop. 6)	oracle fixed bet or fixed grid; persistent δ
Power of the plug-in bet	Empirical	—
Exclusion of persistent synergy $\geq \delta$	Proved (Prop. 11)	grid threshold $\lvert G\rvert/\alpha$
Time-uniform upper bound on average synergy	Proved (Prop. 12)	for the realised forecasters; any data
$\mathrm{Syn}_{\mathrm{aj}} = 0$ under additivity	Proved under (S1)–(S3)	population least squares
Isolation without (S1)–(S3)	False (Prop. 10d)	—
Population null $\Rightarrow H_0^{\mathrm{preq}}$	Not implied (Rem. 3)	—
FWER / FDR across m hypotheses	Proved	arbitrary dependence; e-BH at stopping times
Level under correlated, heavy-tailed, nonstationary nulls	Empirical (Sec. 6.1)	—

6.1 Validity, stress tests and power

Interventional test under continuous monitoring.

On 2000 null streams of length 4000 (balanced 2×2 design, $\theta = 0$) the one-sided e-process rejects in 0.019 of streams (95% CI $[0.014, 0.026]$). A Wald test recomputed after every observation (the optional-stopping failure of Remark 2) rejects in 0.513 $[0.492, 0.535]$. A classical group-sequential design, a Lan–DeMets-spent Pocock or O'Brien–Fleming test with 20 equally spaced looks, rejects in 0.051 $[0.042, 0.061]$ for both boundaries, as designed (boundary constants by Monte Carlo of Brownian look statistics with 400,000 paths, total crossing probability 0.0500). Under an AND mechanism ($\theta = 0.35$) all procedures reject in every stream. The median detection sample size is 390 for the one-sided ANTE process (472 two-sided), 400 for the Pocock design and 1000 for O'Brien–Fleming; the peeking Wald test stops at 102 but with the inflated error just described. A group-sequential design is therefore the right classical comparator and attains similar detection times *when its look schedule is fixed in advance*; the e-process requires no such schedule and remains valid for any stopping rule (Fig. 1). Under the nonstationary nulls of Section 3.1 (contrast switching between -0.2 and 0 in four blocks, and drifting linearly from -0.3 to 0) the one-sided process rejects in 0.001 of streams (CIs up to 0.004), whereas the two-sided process, whose null is false by construction at times when $\theta_t \neq 0$, rejects in 0.583 and 0.955 as it should (Remark 1). The interventional and predictive Type-I figures in this section come from different designs (2000 streams of length 4000 versus 250 streams of length 3000 with 15 looks) and are not directly comparable.

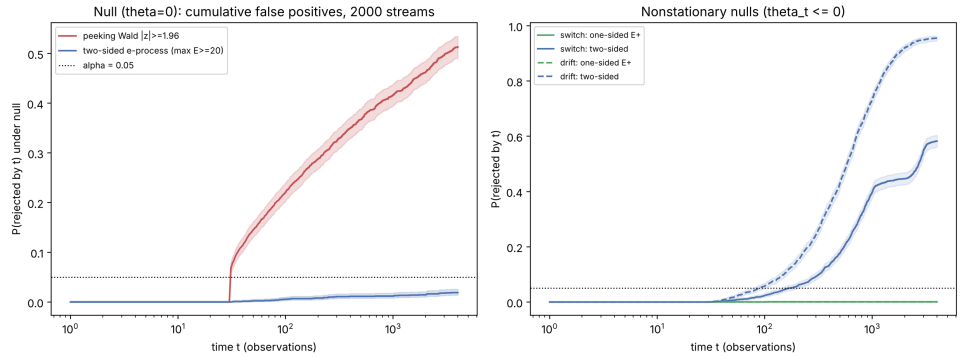


Fig. 1 Interventional test. Left: cumulative false-positive rate under $\theta = 0$ (2000 streams, shaded Wilson bands) for a Wald test recomputed after every observation and the ANTE e-process. Right: under nonstationary nulls with $\theta_t \leq 0$ the one-sided process stays at zero; the two-sided process rejects because its null is false at times

The clipping argument fails; the new score does not.

We reproduce Example 1 by simulation. With $\hat{\sigma} = 1$ and clip level 5, a plug-in bettor on i.i.d. draws from the example law ($\mathbb{E}s = -0.65$) reaches $E \geq 20$ in 0.978 of 2000 streams ($[0.970, 0.983]$, median $\log_{10}$ of the running maximum 8.9), although the raw null is true; with a data-dependent scale (the expanding root mean square) the same law is not harmful (0.004), which is why such a defect can go unseen in simulations. For the new score no such construction exists for the prequential null, because Theorem 8 does not use the shape of the law of s_t. Table 3 and Fig. 2 report ANTE-SG, the four-model score and the original test on twelve VAR nulls ($T = 2500$, 200 replications each) built around the structure in the proof: heavy-tailed skewed innovations, GARCH volatility clustering, source correlation up to 0.99, cubic additive terms outside the feature span with correlated sources, a sign break of the additive coefficients at $T/2$, own-nonlinearity, persistent (AR $= 0.9$) sources, and short (0.98) and long (0.999) forgetting. ANTE-SG's false-rejection rate is at most 0.010 (2/200) in every scenario. The four-model score rejects 0.170 $[0.124, 0.228]$ of nulls under persistent sources (N10) and 0.160 $[0.110, 0.227]$ at a realised source correlation of 0.20 (Section 6.2), although its raw mean score is negative in those cells (about -0.004 and -0.02). The rejections therefore do not come from redundancy, which Proposition 10(b) predicts to make this score negative here; they indicate that the prequential null of the four-model score fails in finite samples for the learned forecasters (Remark 3; we have not isolated the cause), another reason to prefer the add-versus-joint contrast. As an additional check against spurious-regression behaviour, independent AR(1) series with coefficient up to 0.999 (100 replications each) give ANTE-SG false-rejection rates of at most 0.01 (1/100 at $\varphi = 0.97$, none otherwise) and none for the original test. The limits of this evidence appear in Section 6.5 and the discussion: smooth, near-deterministic signals break it.

Power and detection time.

For a bilinear mechanism $Y_t = cA_{t-1}B_{t-1} + 0.2A_{t-1} + 0.3Z_t + \varepsilon_t$ (realised source correlation 0.36, $T = 2500$, 500 replications per c; the figures at family-wise thresholds

Table 3 False-rejection rate (max $E \geq 20$, 200 replications, $T = 2500$) under additive or no-synergy nulls, with Wilson 95% intervals. Scenarios (realised source correlation in brackets): N1 Gaussian (0.36); N2 Student-t_3 and skewed innovations (0.36); N3 t_3 and GARCH (0.36); N4, N5 strongly correlated sources (nominal 0.95, 0.99; realised 0.96, 0.99); N6, N7 additive cubic term outside the feature span (0.60, 0.92); N8 sign break of the additive coefficients at $T/2$; N9 own-nonlinearity (0.60); N10 AR(0.9) sources (0.60); N11, N12 forgetting 0.98, 0.999

Scenario	ANTE-SG (add-vs-joint)	four-model score	original clipped test
N1–N2, N7, N9	0/200 each	0/200 each	0/200 each
N3 (heavy tails + GARCH)	1/200	0/200	0/200
N4 (nominal $\rho = .95$)	1/200	0/200	0/200
N5 (nominal $\rho = .99$)	2/200 [0.003, 0.036]	0/200	0/200
N6, N8, N11, N12	1/200 each	0/200 each	0/200 each
N10 (persistent sources)	0/200	**34/200** [0.124, 0.228]	0/200

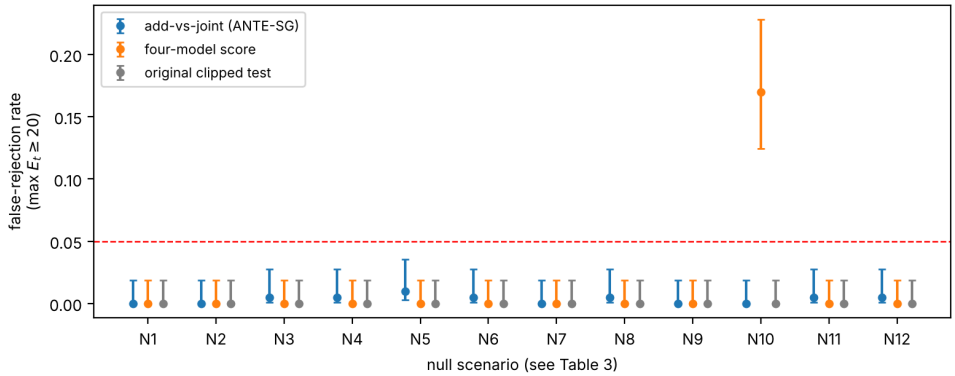


Fig. 2 False-rejection rate of the three procedures under the twelve nulls of Table 3 (Wilson 95% intervals; dashed line: $\alpha = 0.05$)

below use 150 other replications and agree within the intervals), ANTE-SG's power is 0.07 $[0.05, 0.09]$ at $c = 0.1$, 0.78 $[0.74, 0.81]$ at $c = 0.2$ and ≥ 0.998 for $c \geq 0.3$, with false-rejection 0.000 at $c = 0$; the original test reaches only 0.34 at $c = 0.2$ and 0.60–0.69 for $c \geq 0.3$. At the family-wise thresholds the picture is less favourable (150 replications): at $c = 0.2$ power falls from 0.83 at $1/\alpha$ to 0.06 at m/α with $m = 660$ and 0.03 with $m = 4400$; at $c = 0.3$ it is 0.99, 0.77 and 0.61, and at $c = 0.45$ it is 1.00 throughout. The median detection time of ANTE-SG falls from 1346 $(1256, 1438)$ at $c = 0.2$ to 140 $(132, 147)$ at $c = 0.9$ (bootstrap intervals), directionally consistent with the δ^{-2} scaling of Proposition 6 (the effect grows 4.5-fold while the median detection time falls 9.6-fold; the proposition is for an oracle bet and a different score, and we do not claim the bound is tight). For the original test, power dips at the largest c (0.69 $\rightarrow$ 0.63, within Monte Carlo error: the intervals overlap) and latency rises (1095 $\rightarrow$ 1473, outside Monte Carlo error because that test's scale estimator degrades); ANTE-SG's power is monotone within its intervals and its latency non-increasing in $c \geq 0.2$ (Fig. 3). Theorem 5 and Proposition 6 apply to ANTE-SG with ψ_t replaced

by the normalised score u_t ($c = 1$) and a fixed bet; the plug-in bet is evaluated here only empirically.

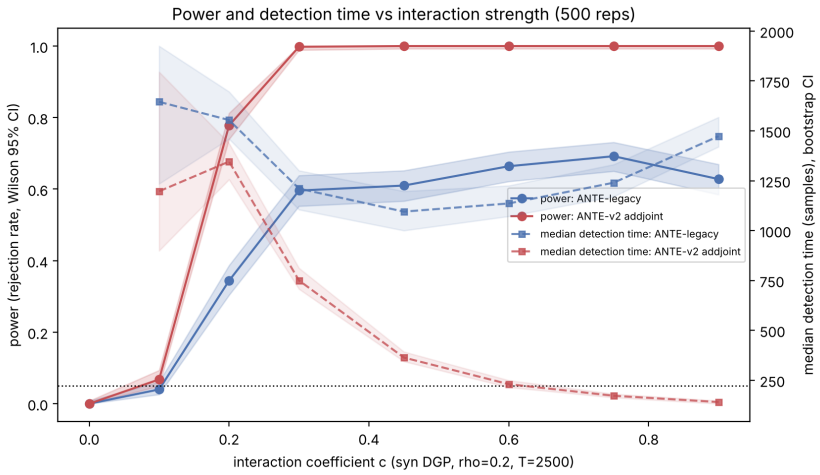


Fig. 3 Power (solid, left axis) and median detection time (dashed, right axis) versus the interaction coefficient, with Wilson and bootstrap bands (500 replications per point), for the original test and ANTE-SG

6.2 Correlated sources and nonlinear alternatives

A natural question is what happens when sources are strongly correlated while the mechanism is additive. We vary the source correlation (realised correlation 0.20 to 0.99) for an additive mechanism and for an interaction of strength 0.9 (150 replications per cell). ANTE-SG's false-rejection rate is 0.000 in all seven cells ($[0, 0.025]$). The mean raw four-model score decreases from -0.02 at correlation 0.2 to -1.04 at 0.99 on the additive mechanism, as the redundancy term predicts, whereas the add-versus-joint score stays between -0.04 and -0.007 (Fig. 4). The cost of isolating the interaction is power when the two sources become nearly collinear: ANTE-SG's power against the interaction is 1.00 up to correlation 0.68, 0.97 $[0.92, 0.99]$ at 0.84, 0.19 $[0.13, 0.26]$ at 0.92, 0.007 at 0.96 and 0 at 0.99. When $A \approx B$ the cross-product AB is nearly collinear with the own-quadratic features A^2, B^2 and the interaction is not identifiable; this is a property of the problem, not an artefact, and it bounds what any add-versus-joint test can claim for near-collinear assets (for example for two equity sector funds). The four-model score loses power sooner (0.013 at correlation 0.68) because its redundancy term turns negative, and the original test is the least powerful throughout (0.73 at correlation 0.2, 0.31 at 0.44, 0 from 0.68).

The test is only as nonparametric as its predictor class.

The dictionary used above (own linear and squared lags plus cross-products) detects a bilinear and a bounded $tanh \times tanh$ or sin interaction (100 replications, power 1.00

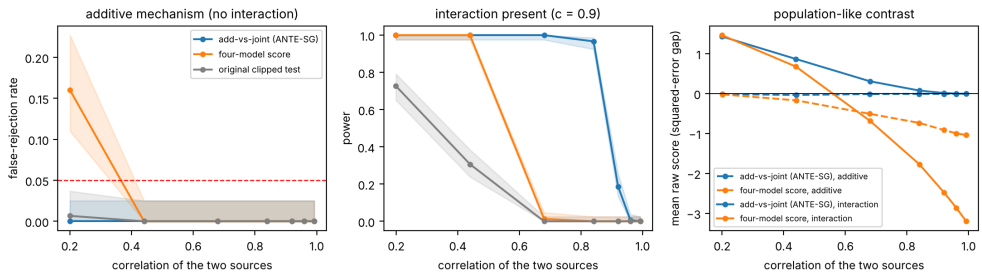


Fig. 4 Correlated sources. Left: false-rejection rate under an additive mechanism. Centre: power against an interaction of strength 0.9. Right: mean raw score (squared-error gap); the four-model score becomes strongly negative with redundancy while the add-versus-joint score stays near zero. Bands: Wilson 95% intervals (150 replications)

each) because their projections onto the cross-product are strong, but it fails on $Y_t = 0.9A_{t-1}^2 B_{t-1} + \ldots$, an interaction outside its span: power 0.03 $[0.01, 0.08]$ at $T = 2500$ and again at $T = 5000$. Replacing the dictionary by a random-Fourier-feature map (Rahimi and Recht 2007) (fifteen features per source and fifteen for the pair, with fixed seed-independent frequencies) detects it with power 1.00 $[0.96, 1.00]$ at both sample sizes, while its false-rejection rate on the additive null remains 0.00 $[0, 0.04]$. The framework (the betting construction and Theorem 8) accommodates any predictable forecaster pair; the RLS-on-polynomials instantiation is one choice, and we describe it as a "model-based prequential time-series implementation of an anytime-valid testing framework" rather than as a nonparametric test.

6.3 Benchmark against batch synergy estimators

SURD (Martínez-Sánchez et al. 2024), PEID (Yang et al. 2026), Williams–Beer PID (Williams and Beer 2010), Gaussian interaction information (McGill 1954) and O-information (Rosas et al. 2019) are *estimators* of information-theoretic synergy, a different estimand from the predictive contrast tested here. We therefore report the comparison as one of complementary tools. Each baseline was first checked against its own source: our PEID implementation returns 1.00 for XOR and 0.19 for AND (published value 0.189), PID and SURD agree on XOR (1.00) and give 0.50 on AND, and all return ≈ 0 on a redundant copy and on independent noise. Settings (not tuned per method): SURD with six equal-mass bins per variable on the lagged triple (A_{t-1}, B_{t-1}, Y_t) through the authors' released code; PEID with polynomial-lifted affine-Gaussian mutual information under an independent maximum-entropy intervention; interaction information the Gaussian co-information of the same triple; for the monitoring comparison a permutation test on interaction information with 50 permutations at 15 equally spaced looks. Detection AUC uses 100 replications of each of three DGPs (additive, own-nonlinear, synergistic; $T = 3000$, per-replication seeds).

On detection of the DGP built for the add-versus-joint contrast, batch SURD and PEID are as accurate as ANTE-SG (AUC 1.00 each; the original test, 0.992, and interaction information, 0.865, are below), so ANTE-SG does not dominate the batch

Table 4 Detection AUC (synergistic versus additive and own-nonlinear DGPs; 100 replications per DGP, $T = 3000$; bootstrap 95% intervals) and behaviour under repeated looks on an additive null (250 replications, 15 looks; Wilson intervals) and power on the synergistic DGP (100 replications)

Procedure	AUC [95% CI]	Type-I at 15 looks	power (synergistic)
ANTE-SG (add-vs-joint)	1.000 [1.000, 1.000]	0/250 [0, 0.015]	1.00 [0.96, 1.00]
Original clipped test	0.992 [0.983, 0.998]	0.020 [0.009, 0.046]	0.81 [0.72, 0.88]
SURD	1.000 [1.000, 1.000]	— (point estimate)	—
PEID	1.000 [1.000, 1.000]	— (point estimate)	—
Interaction information	0.865 [0.824, 0.906]	0.116 [0.082, 0.162] (permutation)	0.63 [0.53, 0.72]
Group-sequential Pocock (raw score)	—	0/250 [0, 0.015]	0.99 [0.95, 1.00]
Group-sequential O'Brien–Fleming (raw score)	—	0/250 [0, 0.015]	1.00 [0.96, 1.00]

estimators and is not claimed to; they answer a different question and return a value without an error guarantee under repeated looks, which is what the permutation-based entry measures. The group-sequential tests applied to the raw per-step synergy score are valid at their 15 pre-planned looks and powerful, with nominal errors below 0.05 because their variance is estimated and the raw score is conservative; they are the appropriate classical reference, and ANTE-SG's advantage over them is not accuracy at a pre-planned look but validity at *any* stopping time. Gaussian PID and the other decompositions assign positive synergy to $Y = A + B$ because their estimand is "information about Y available only from A and B jointly"; we do not describe this as an error. The SURD/PEID rows for additive mechanisms are therefore a difference in estimand, not a failure. The distributional Granger test of Jha (2026) is a sequential test of a single source and cannot express a group.

6.4 Positive controls on real data

Contemporaneous portfolio variance (explicit target).

The realised variance of a two-asset equal-weight portfolio, $RV_t = (0.5r_{A,t} + 0.5r_{B,t})^2$, is a deterministic function of the two contemporaneous returns, so its super-additivity ($2w_A w_B \sigma_{AB}$) is mechanical. It is a check that the contemporaneous variant can recover an interaction that is known to be there, not an example of group causality: we do not call it causal synergy, and we separate three notions (contemporaneous statistical interaction; predictive group synergy; causal/interventional synergy) throughout. On 14 asset pairs (listed in the repository), RV_t as target, contemporaneous features and no conditioner, ANTE-SG rejects in all 14 at level 0.05 and in 13 at the family-wise threshold $14/\alpha$ (the remaining pair, XLF/XLI, has correlation 0.86). The lagged counterpart is not significant for any pair at the family-wise threshold (1/14 at the raw level). Two features of this pattern need explanation, and both follow from the target definition. The near-collinear within-equity pairs have the weakest signal ($\log_{10} E$ of 5.4 to 8.9 for XLK/XLI, XLE/XLF, XLE/XLB; 1.7 for XLF/XLI) because the regression of the cross product $r_A r_B$ on the own features $(1, r_A, r_B, r_A^2, r_B^2)$ has $R^2 = 0.955$–0.995 for them, so the cross term is almost fully explained by the own-quadratic terms

already in the additive model; for the cross-asset pairs that R^2 is 0.03–0.45. Equity–bond pairs have negative covariance (for example XLF/TLT, corr -0.26) and are detected strongly: the test measures the *predictive gain* from the cross term, which does not depend on the sign of the covariance, so negative covariance does not make a pair "sub-additive" in this sense. Capital saturates the 10^{15} cap for most cross-asset pairs.

Semi-synthetic power on real noise.

To measure power on genuine financial noise, we inject a known lagged interaction into real series: $Y'_t = z(C)_t + \beta\, z(A)_{t-1} z(B)_{t-1}$ for 40 random real triples of each of two datasets (daily stock returns conditioned on SPY; daily ETF volatility conditioned on |SPY|), re-standardised predictably. At $\beta = 0$ (Type-I on real noise) the single-hypothesis rejection rate is 0.025 and 0.050 (40 triples each), and no triple reaches a family-wise threshold. Power is low even for large injected terms (Fig. 5): at $\beta = 0.5$, an injected term that carries on average 40% (35%) of the variance of Y', power is 0.43 (0.25) at the single-hypothesis threshold, 0.28 (0.13) at the $m = 660$ family-wise threshold and 0.23 (0.13) at $m = 4400$; 0.8 power is not reached on the grid. A natural suspicion is that the truncated loss and the feature winsorisation, which make the test robust, are responsible, because the injected term (a product of two heavy-tailed standardised returns) concentrates its variance in a few extreme days. We tested this by relaxing both (truncation level $c = 6$, no winsorisation) on the same triples: power at the family-wise thresholds is still at most 0.075 (0.10) at $\beta = 0.5$, and the single-hypothesis power is 0.33 and 0.38 with the injected term carrying 54% and 68% of the variance (the comparison is not variance-matched), while the rejection rate at $\beta = 0$, a true null on real noise, rises to 0.075 and 0.20. Relaxing the robustness therefore brings no material gain at the family-wise thresholds and costs validity on real data; we could not isolate the cause of the low power; the practical consequence for interpretation is that a family-wise non-rejection does *not* show that no interaction exists. Section 6.5 accordingly reports bounds on the average synergy of the learned forecasters.

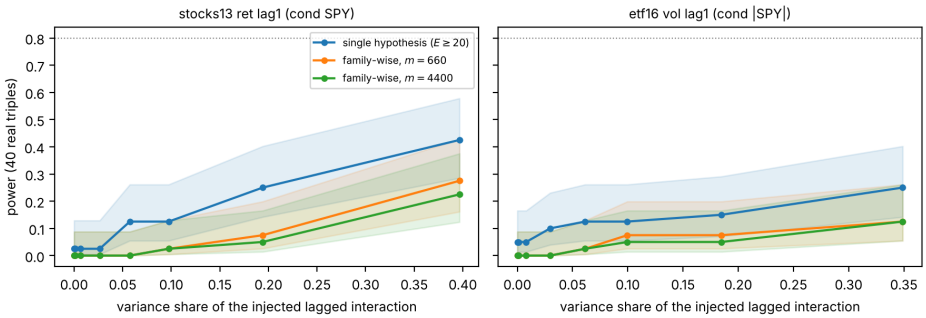


Fig. 5 Power on real financial noise when a known lagged interaction is injected (40 real triples per panel): single-hypothesis threshold and family-wise thresholds for $m = 660$ and $m = 4400$ hypotheses (Wilson bands)

Turbulence data are not a valid control.

The SURD turbulence signals (three coarse scales and the finest scale of an energy cascade) look like an ideal real-data positive control for a three-way mechanism, but they fail a placebo check. Using a null that has the same autocorrelation as the data but is independent of the sources, phase-randomised surrogates of the finest-scale signal, both the clipped-score test and ANTE-SG "detect" on 20/20 surrogates (median $\log_{10}$ final evidence 11.4 for the clipped test, against 4.0 on the real series; median $\log_{10}$ running-maximum evidence 5.2 for ANTE-SG), while ANTE-SG does not reject on the real series ($\log_{10} E_{\max} = 0.95$). The signals are smooth and nearly deterministic (lag-one autocorrelation 0.9991–0.9995, and 0.96–0.97 after differencing) and the learned forecasters violate the prequential null in long stretches (Remark 3), so these data cannot validate either procedure and we do not use them as evidence. By contrast, independent AR(1) series up to 0.999 do not produce false rejections (Section 6.1); the failure is specific to smooth, near-deterministic dynamics. The k-way test (Corollary 9) is validated on synthetic data: a pure three-way mechanism $Y = 0.9ABC$ or $0.5ABC$ is detected in 100/100 replications ($[0.96, 1.00]$) and a three-source additive null is not rejected (0/100, $[0, 0.04]$).

6.5 A fully specified search of financial data

Scope and accounting.

All data are public and listed in Table 5. A hypothesis is a target asset C, an unordered pair $\{A, B\}$ of other assets and a setting, tested conditionally on the market factor; the number of hypotheses per panel and setting is $N'\binom{N'-1}{2}$ with N' the number of assets other than the conditioner. The search space is written down in `experiments/revision/results/r5_search_space.json` before the scan ran. It has three target types: returns, volatility ($|r|$), and stress indicators (the indicator that $|r_t|$ exceeds the expanding 90% quantile of its own past); and four structures, chosen in advance in the spirit of pre-specified lag grids (Choi 2026): lag 1, lag 5 (daily panels), weekly aggregation (5-day sums, lag 1) and a contemporaneous variant (sources at time t; this is labelled "contemporaneous interaction", not Granger prediction). There are 43 settings ("families") and 27,027 hypotheses in total; after removing the 132 tuples that appear identically in the ETF and stock panels (shared assets GLD, SPY, TLT, USO, VIX) about 26,900 are distinct. The search is *not* pre-registered. Earlier exploratory scans of related questions (volatility, stress and crypto targets, other frequencies) preceded the grid, which was fixed before it was run but designed after those results were known; the assets and panels were chosen by the author as a standard set, not by selecting on outcomes, and no hypothesis was dropped. The tuning constants ($\rho = 0.995$, $\gamma = 0.9$, warm-up 50) were set in that earlier work on the same data, and the truncation level $c = 3$ and winsorisation at ± 5 before this scan but after seeing those earlier results, so the anytime-valid guarantee, which requires constants fixed independently of the series under test, holds for them only as a matter of design discipline. We therefore report each family with its own family-wise threshold m/α and also a global threshold with $m = 27{,}027$ ($\log_{10}(m/\alpha) = 5.73$); e-values

are combined by e-Bonferroni, which is valid under arbitrary dependence among the overlapping hypotheses.

Table 5 Datasets (all public). Panels used in the search: the number of hypotheses is given for each (target, structure) setting

Panel	Content	Size	Hypotheses / setting
etf16	16 daily ETFs and macro assets (2016–26), conditioner SPY	$T = 2512$	1365
stocks13	13 daily stocks and assets, conditioner SPY	$T = 2512$	660
world6	6 daily equity indices, conditioner US S&P	$T = 2138$	30
hourly12	12 hourly stocks and assets, conditioner SPY	$T = 3328$	495
crypto10	10 hourly crypto assets, conditioner BTC	$T = 16{,}761$	252
Synthetic VAR / interventional	ground truth	2500–5000 steps	simulations
Turbulence signals	four scale-resolved signals (not used as evidence)	21,760	—

The volatility index VIX (in the ETF and stock panels) is an index, not a traded asset; its log changes are used as a series and read as changes in implied volatility.

Results.

Table 6 summarises the search and Fig. 6 shows the distribution of running-maximum evidence per hypothesis. For *returns and volatility in daily and weekly panels*, 26 families containing 18,360 hypotheses give 143 raw rejections (0.8%, below the nominal 5% as expected for a conservative e-process) and, at the family-wise threshold of their own family, *no* rejection in any of the 22 daily and weekly families (etf16: 0/10,920; stocks13: 0/5280; world6: 0/180), for lag 1, lag 5, weekly and contemporaneous structures alike. The single family-wise rejection among stock and ETF returns and volatility is one of 495 hourly stock-volatility hypotheses in the contemporaneous structure ($\log_{10} E = 4.24$ against a threshold of 4.00); it does not survive the global threshold. The largest running-maximum evidence in any daily or weekly returns/volatility family is $\log_{10} E = 3.68$. The median final $\log_{10} E$ is -0.24 (a typical bet on synergy loses about 40% of its stake).

Stress indicators and hourly crypto behave differently. For stress indicators in daily and hourly stock panels, 1588 of 7155 hypotheses (22%) reject at the raw level and 543 at their family-wise thresholds; for hourly crypto, 599 of 1512 (40%) and 376. e-BH on final e-values, pooled within each group (FDR 0.1 or 0.05), makes no discovery for returns and volatility, 212 and 195 discoveries for stress indicators and 348 and 339 for crypto (513 and 491 pooled over all 27,027 hypotheses). These counts are not evidence of synergy, for the reason given in the next paragraph: the placebo test shows that the test over-rejects on exactly these targets.

Table 6 Search summary (ANTE-SG, add-vs-joint score). m: hypotheses; raw: running maximum ≥ 20; fw: running maximum $\geq m/\alpha$ within the family. Groups pool settings (lag 1, lag 5, weekly, contemporaneous)

Group	families	m	raw	fw
Returns, volatility: daily/weekly panels	22	16,380	99	0
Returns, volatility: hourly stocks	4	1980	44	1
Stress indicators: stocks and ETFs	11	7155	1588	543
Crypto, hourly (all targets)	6	1512	599	376
All	43	27,027	2330	920*

*Sum of per-family thresholds; 657 hypotheses cross the single global threshold m/α with $m = 27{,}027$.

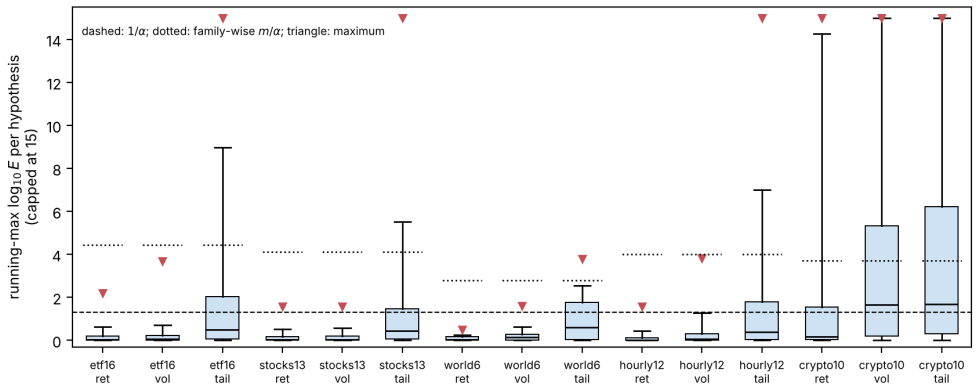


Fig. 6 Running-maximum evidence $\log_{10} E$ per hypothesis for the lag-1 families of each panel and target (box: quartiles, whiskers: 5th–95th percentile, triangle: maximum, dotted line: family-wise threshold, dashed: $1/\alpha$). Returns and volatility stay below the family-wise thresholds in daily panels; stress indicators and crypto do not

Null calibration on the real data.

A simulated null says little about how the test behaves on real series. We therefore calibrate on the data themselves: for a random sample of hypotheses, the target is circularly shifted by at least a year (a quarter, for weekly data) relative to sources and conditioner, which preserves its marginal law and autocorrelation but destroys every cross-dependence, so every null is true. For daily stock volatility, daily ETF returns, hourly stock volatility and hourly stock returns (250, 250, 250 and 200 hypotheses), the distribution of the running maximum under the placebo coincides with the real one up to the extreme tail (Fig. 7); raw rejections are 2/250 (real) versus 0/250 (placebo) for stock volatility, 2/250 versus 0/250 for ETF returns, 10/250 versus 9/250 for hourly volatility (placebo rate 0.036 $[0.019, 0.067]$) and 1/200 versus 1/200 for hourly returns. Across nine further returns-and-volatility families (lag 5, weekly, contemporaneous and lag-1 variants, 150 hypotheses each) the placebo rejects in at most 2/150 hypotheses (0.013) against at most 4/150 for the real series. The real evidence in these settings

therefore sits on the placebo null, which is a more convincing calibration than comparison with a simulated null. It also explains why the median final $\log_{10} E$ is negative: with the present test the simulated null median is -0.15, the placebo median -0.20 to -0.26 and the real financial median -0.27 (150 real triples). The drift concentrates in the first third of the sample (median contribution -0.17 for finance, -0.11 for the simulated null), where the forgetting learners are still estimating the larger model, as the out-of-sample penalty of Section 5.4 predicts, and is not driven by winning bets that later reverse (a running maximum above 0.3 followed by a final value below -0.3 occurs for 13% of finance triples and 8%–11% of simulated nulls).

Where the calibration fails. The same placebo applied to the families that do reject shows that the test is not calibrated there (Fig. 7). With a true null, hourly crypto returns reject in 76/200 hypotheses (0.38, [0.32, 0.45]; real data 53/200) and hourly crypto volatility in 79/200 (0.40; real 108/200); binary stress indicators reject in 28/200 (0.14) for the ETF panel, 19/200 (0.095) for the stock panel and 56/200 (0.28) for hourly stocks (real data: 68/200, 56/200 and 65/200). For these targets the prequential null is evidently violated by the learned forecasters; we conjecture that the cause is the binary or extremely heavy-tailed nature of the series, with strong volatility clustering and a nonlinear link between the cross-product features and the target, but we have not tested it. In any case a rejection does not indicate an $A \times B$ interaction. For daily stress indicators the real rejection rates exceed the placebo rates (0.34 versus 0.14; 0.28 versus 0.095), which suggests some genuine cross-sectional dependence in tail events beyond the test's own bias, but we cannot separate it from a misspecified forecaster.

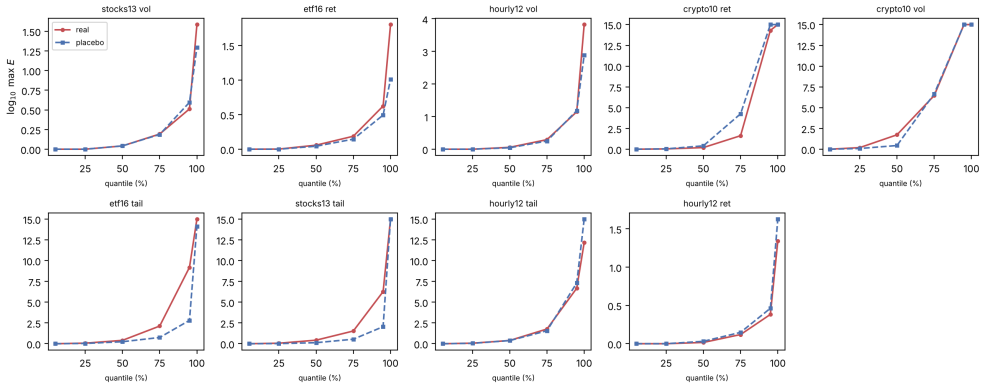


Fig. 7 Real versus placebo (circularly shifted target) distribution of the running-maximum evidence for random samples of hypotheses; coinciding curves indicate that real data behave like the true null in the setting

Truncation share.

For the standardised series of the search, 1.5–2.3% of the observations lie beyond three predictable standard deviations but carry 34–63% of the sum of squares (36%–54%

for daily returns and volatility, 63% for hourly stock volatility), so losses truncated at $c = 3$ discard a large share of the variance; this is why the loss units of Section 5.5 are not untruncated R^2.

How much is excluded.

The persistent-alternative e-process of Proposition 11 is uninformative on these data, because the learners often agree to within the smallest δ on the grid (0.0005) and the process then wins mechanically. We therefore use the time-uniform bound on the *running-average* conditional synergy of Proposition 12 (loss units; $K = 22$ grid points; level 0.05 divided over the hypotheses of a family; Table B1 in Appendix B). For the lag-1 and contemporaneous families of daily returns and volatility, the median hypothesis excludes an average synergy above 0.057–0.122 loss units, but the least well-bounded hypothesis of a family only excludes values above 0.11–2.15; for lag-5 and weekly families, where the forecasters have more regressors or fewer observations, the median bound is 0.22–0.75 and the worst case is 0.77–1.91 (weekly) and 6.4–19.6 (lag 5). The realised mean score is negative in every family (median -0.005 to -0.034 per step), as the out-of-sample penalty predicts. Because the bound uses the worst-case variance proxy b_s^2, it is conservative, and it is weak against magnitudes that are plausible for daily returns (a gain of order 1% of the variance): for a typical hypothesis the data exclude average synergies above roughly 6–12% of the (truncated) target variance per step, while the simultaneous guarantee for a whole family is the maximum bound (0.11–2.15 at lag 1 and contemporaneous), and neither excludes smaller effects.

Crises.

Running e-processes are plotted across whole families in Fig. 8, so that evidence that rises in a crisis and then decays is visible, which a final-value summary would hide. In the 660 stock-volatility and 1365 ETF-return families the maximum over hypotheses of $\log_{10} E_t$ never approaches the family-wise thresholds (4.1 and 4.4): its peak is 1.59 (January 2017) and 2.20 (5 February 2020, *before* the COVID sell-off, after which it decays), and during the three windows examined (the 2018 volatility spike, February–June 2020, and 2022) the median over hypotheses of the within-window maximum is between -0.10 and -0.22, with only 1.2%–2.2% of hypotheses exceeding $\log_{10} E = 0.3$ at any time in a window. The cross-sectional median drifts to about -0.26 to -0.30 and the 95th percentile stays at zero (the bet is switched off).

Truncation level and winsorisation.

For 200 random hypotheses from each of daily stock volatility and daily ETF returns, varying the truncation level ($c = 2, 3, 5$) and removing the feature winsorisation changes the raw rejection count by at most two hypotheses (1–2/200 at $c = 2$, 0/200 for $c = 5$ or without winsorisation, 1/200 at the default) and the median final $\log_{10} E$ stays between -0.15 and -0.33; the conclusions for daily returns and volatility are therefore not sensitive to these choices.

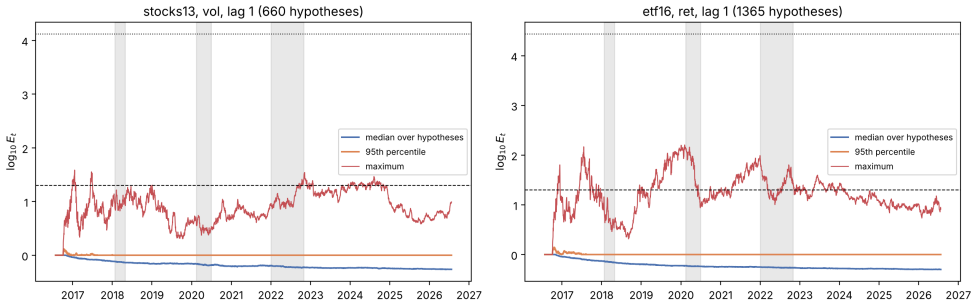


Fig. 8 Running evidence $\log_{10} E_t$ across all hypotheses of two families (median, 95th percentile, maximum). Shaded: the 2018 volatility spike, the 2020 COVID window and 2022; dashed: $1/\alpha$; dotted: family-wise threshold

Forgetting factor and the Clark–West diagnostic.

With forgetting factors from 0.98 to 0.999 the false-rejection rate is at most 0.010 in every simulated null cell (100 replications each; Wilson upper limits up to 0.054) and the real-data rejection rate at most 0.047 (150 financial triples, $\rho = 0.98$), and the median final $\log_{10} E$ stays between -0.04 and -0.27; $\rho = 0.98$ makes the learners unstable (the Clark–West adjustment term explodes) and is not recommended. The Clark–West-adjusted mean score is 0.03–0.13 for $\rho \in [0.99, 0.999]$ on both simulated and financial data, which gives the size of the noise floor; as stated in Section 5.4 we do not use it for inference.

The pairwise layer.

Billio et al. (2012) measure connectedness through pairwise Granger networks. On the same panels, the fraction of significant ordered pairs (Bonferroni at 0.05) is 0.20 for ETF returns and 0.78 for ETF volatility at lag 1 (0.10 and 0.61 for stocks), and a rolling 252-day window (raw 5% level) gives a density of up to 0.55 (ETFs) and 0.40 (stocks) around the 2020 crash against a median of about 0.1. Daily markets therefore have a strong pairwise layer, particularly in volatility. For 660 stock-volatility triples, 145 have both single-source tests significant after conditioning on SPY (29 after Bonferroni), but only 4 reach a raw ANTE-SG rejection and none a family-wise one: the pairwise structure is not accompanied by detectable super-additive structure.

6.6 Interpretation

We summarise what the evidence supports, keeping the wording to the searched space. In daily and weekly returns and volatility of the panels examined, for lag 1, lag 5, weekly and contemporaneous structures, the ANTE-SG test finds no statistically significant super-additive predictive group effect at the family-wise level, and the real-data evidence has the distribution of a placebo null. Where significant evidence does appear (stress indicators; hourly crypto), a placebo test shows that the test rejects a true null in 10–40% of hypotheses for these targets, so the rejections are not interpretable as synergy and the test would need a different forecaster or score (binary or

heavy-tailed targets) to be usable there; We do not claim that synergy is absent from financial markets in general, and not even from the settings searched, for three reasons developed above: the test has low power on real financial noise even for large injected interactions (semi-synthetic power), the predictor class limits what can be detected, and non-rejection is a statement about these online forecasters (Remark 3).

7 Discussion and conclusions

ANTE reframes synergy detection from estimation to *testing* and from fixed-sample to *anytime-valid* inference. The methodological result is a pair of tests with time-uniform error control: an interventional test under a known randomised design whose null is sequential and therefore allows nonstationarity, and a predictive test for time series whose null compares two learned forecasters and whose validity proof is exact (Theorem 8) once the score is built from a truncated loss and a bound that is known before each observation. The cost is low power on real financial noise, which we quantified by injection (Section 6.4) and whose cause we could not isolate (relaxing the truncation does not cure it); a second cost, common to every anytime-valid procedure, is lower power than an oracle fixed-sample test at a single pre-committed sample size.

Arguments that do not hold.

Three tempting arguments are avoided here. (i) A validity proof for the group score by clipping fails, because clipping does not in general preserve the sign of the conditional mean (Example 1). (ii) Isolation of an interaction from joint dependence does not follow from additivity of the mechanism: it needs orthogonality, which holds under independence or span conditions and can fail when sources are dependent and the mechanism lies outside the feature span (Proposition 10), and the four-model contrast is contaminated by redundancy and suppression. (iii) A real-data positive or negative control is only informative if the test has been calibrated against a placebo null on the same kind of data: smooth near-deterministic signals (turbulence), binary stress indicators and hourly crypto series show that the test can reject a true null there.

Limitations.

(1) A rejection means that the joint forecaster had a persistently positive conditional advantage over the additive one (Remark 3); reading it as an irreducible $A \times B$ interaction needs the conditions of Proposition 10, which cannot be checked in data, and reading it as causal needs an interventional design. (2) The prequential null is not implied by zero population synergy, so Type-I control for population-level nulls is established by simulation and placebo calibration, not by theorem; the simulations include heavy tails, GARCH, nonstationarity and correlated sources but are not exhaustive, and the test fails on smooth, near-deterministic signals such as the turbulence data. (3) Power: the test has low power for lagged interactions between heavy-tailed returns (which relaxing the truncation does not cure), loses power as the two sources become nearly collinear (the interaction is then not identifiable), and its detectable alternatives depend on the dictionary; a random-feature dictionary removes one blind spot but the choice of dictionary is fixed in advance. For the implemented

plug-in bet we prove validity only. (4) Families are not pre-registered and overlap (shared assets, repeated targets), so the e-Bonferroni thresholds, though valid under arbitrary dependence, are conservative. (5) Conclusions are specific to the panels, targets, lag structures and sample periods searched; for hourly crypto data and binary stress indicators the test is not calibrated (Section 6.5), so the significant evidence there cannot be interpreted. (6) The augmented (AIPW) interventional score with estimated propensities is an extension without a finite-sample guarantee.

Future work.

Kernel or neural forecasters in the same betting scaffold, together with the sequential kernelised tests of Podkopaev et al. (2023); anytime-valid conditional-independence tests (Grünwald et al. 2024) to condition on confounders and move from predictive to causal readings; a sequential comparison with regret guarantees for the plug-in bet; and an information-theoretic instantiation in which a permutation-recentred PEID increment is bet on.

Conclusion.

We introduced ANTE, an anytime-valid sequential test of group synergy with an interventional and a predictive formulation. It controls Type-I error uniformly over all stopping times for the sequential null of the interventional part and for the forecaster-comparison null of the predictive part (Theorems 2 and 8); for population-level nulls in real data, control is established only by simulation and placebo calibration. The test has power one and a high-probability detection-time bound for an oracle bet (Theorem 5, Proposition 6), extends to k-way groups (Corollary 9), and comes with an exclusion e-process, a time-uniform bound on average synergy and a precise account of what a rejection means (Proposition 10). Empirically it holds its level under continuous monitoring and under a range of simulated stress nulls and, on real data, in daily returns and volatility (but not for binary stress indicators or hourly crypto, where it is miscalibrated), and in a fully specified search of financial data it finds no statistically significant super-additive predictive group effect in daily and weekly returns and volatility, with real-data evidence that matches a placebo null. The methodological result and the empirical finding are separate: the first is a theorem and simulation record; the second is a finding about the specific panels searched.

Supplementary information. Code, data and scripts that reproduce every figure and table are available in the public repository (see Declarations).

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Declarations

Ethical approval

Not applicable. This study uses only publicly available market and turbulence (fluid-dynamics) data and synthetic simulations. It involves no human participants, human data or animals, and no ethical approval was required.

Consent to participate

Not applicable. The study involved no human participants.

Consent to publish

Not applicable. The manuscript contains no individual person's data, images or details that would require consent for publication.

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Competing interests

The author has no competing interests to declare that are relevant to the content of this article.

Data availability

All datasets are public. Financial series were obtained from the Yahoo Finance chart API and the turbulence energy-cascade signals (used only in a placebo check) from the released SURD repository (Martínez-Sánchez et al. 2024). Downloaders and processed CSV files are included in the project repository, `github.com/vinhqdang/REIGN_Regime-Enhanced-Intelligent-Granger-Network`, in the subdirectory `sequential-synergy-causality/`.

Code availability

All source code (the ANTE and ANTE-SG implementations including the revised test `src/ante_v2.py`, the re-implemented baselines, the experiment scripts under `experiments/revision/`, and the figure-generation code) is openly released at `github.com/vinhqdang/REIGN_Regime-Enhanced-Intelligent-Granger-Network`, under `sequential-synergy-causality/`. The experiments were run with Python 3.13.16 using NumPy 2.5.3, SciPy 1.18.1, pandas 3.0.5 and Matplotlib 3.11.2 on a four-core Linux container; all random number generators are seeded, and the seeds are given in the scripts.

Author contributions

Q.-V. Dang is the sole author and conceived the method, developed the theory, implemented the algorithms and experiments, and wrote the manuscript.

Appendix A Proofs and supporting results

Throughout, "predictable" means $\mathcal{F}_{t-1}$-measurable, $\gamma < 1$, and Ville's inequality is used in the form: if $(M_t)_{t\geq 0}$ is a nonnegative supermartingale with $\mathbb{E}M_0 \leq 1$ then $\mathbb{P}(\exists t : M_t \geq 1/\alpha) \leq \alpha$.

A.1 Proofs for Part A

Proof of Lemma 1 Write $\psi_t = \sum_{a,b} c(a,b)\mathbb{1}\{A_t = a, B_t = b\}C_t(a,b)/\pi_t(a,b)$. Condition on $(\mathcal{H}_t, \{C_t(\cdot)\})$. By sequential randomisation each term has conditional mean $c(a,b)C_t(a,b)$, because π_t is $\mathcal{H}_t$-measurable. Taking $\mathbb{E}[\cdot \mid \mathcal{H}_t]$ gives $\theta_t(X_t)$, and the tower property gives $\mathbb{E}[\psi_t \mid \mathcal{F}_{t-1}] = \theta_t$. The bound follows from $0 \le C_t \le 1$ and positivity. □

Proof of Theorem 2 and Corollary 3 Each factor $1+\lambda_t\psi_t \ge 1-\gamma > 0$, so $E_t \in (0, (1+\gamma)^t]$ is integrable. Since λ_t is predictable, $\mathbb{E}[E_t \mid \mathcal{F}_{t-1}] = E_{t-1}(1+\lambda_t\theta_t)$. Under $H_0^{\le}$ and $\lambda_t \ge 0$ this is at most E_{t-1}. Under $H_0^{=}$ it equals E_{t-1}, and an average of two martingales for the same filtration is a martingale. Ville's inequality gives the crossing bound, and optional stopping for nonnegative supermartingales gives $\mathbb{E}E_\tau \le 1$ for every stopping time. □

Proof of Lemma 4 Put $q(x) = (x - \log(1+x))/x^2 = \int_0^1 v/(1+xv)\,dv$ for $x > -1$; q is strictly decreasing, so $q(x) \le q(-\gamma) = C_\gamma$ on $[-\gamma, \infty)$, with equality at $-\gamma$. Finally $q(-\gamma) = \int_0^1 v/(1-\gamma v)dv \le 1/(2(1-\gamma))$. □

Proof of Theorem 5 Let $W_s = \log(1+\lambda\psi_s)$. Since $|\lambda\psi_s| \le \bar{\gamma}$, Lemma 4 gives $\mathbb{E}[W_s \mid \mathcal{F}_{s-1}] \ge \lambda\delta - C_{\bar{\gamma}}\lambda^2c^2 = g_0$. The differences $D_s = W_s - \mathbb{E}[W_s \mid \mathcal{F}_{s-1}]$ are martingale differences bounded by $\ell_\lambda = \log\frac{1+\lambda c}{1-\lambda c}$, so $\sum_s \mathbb{E}D_s^2/s^2 < \infty$ and the martingale strong law gives $t^{-1}\sum_{s\le t} D_s \to 0$ a.s. Hence $\liminf t^{-1}\log E_t \ge g_0$ a.s. The statement for the cap-based Kelly bet follows by substituting $\lambda^\star = \delta(1-\gamma)/c^2$ and the bound $C_{\bar{\gamma}} \le 1/(2(1-\gamma))$. (Almost-sure *convergence* of $t^{-1}\log E_t$ would need an ergodicity assumption; the lim inf statement does not.) □

Proof of Proposition 6 $\{\tau_\alpha > t\} \subseteq \{\log E_t < \log(1/\alpha)\}$ and $\log E_t \ge tg_0 + M_t$ with $M_t = \sum_{s\le t} D_s$. Each D_s lies in a predictable interval of length ℓ_λ, so Azuma–Hoeffding (Azuma 1967) gives $\mathbb{P}(M_t \le -\epsilon) \le \exp(-2\epsilon^2/(t\ell_\lambda^2))$; take $\epsilon = tg_0 - \log(1/\alpha) > 0$. For $t \ge 2\log(1/\alpha)/g_0$ the exponent is at least $tg_0^2/(2\ell_\lambda^2)$, which exceeds $\log(1/\beta)$ once $t \ge 2\ell_\lambda^2\log(1/\beta)/g_0^2$. (Using the range at the cap, $\log\frac{1+\gamma}{1-\gamma}$, in place of ℓ_λ is valid but gives a bound of order δ^{-4}.) □

Proof of Proposition 7 An average of supermartingales for one filtration is a supermartingale; a sum of nonnegative terms is at least its largest term, so $E_t^{\mathrm{mix}} \ge K^{-1}\max_j E_t(\lambda_j)$. On the geometric grid, g_0 is a concave quadratic with maximiser $\lambda^\star$, so some grid point in $[\lambda^\star/2, \lambda^\star]$ has $g_0 \ge \frac{3}{4}g_0(\lambda^\star)$. □

Known-design AIPW.

Conditioning as in Lemma 1, the weighted residual term of ϕ_t has conditional mean $\sum c(a,b)(\mathbb{E}[C_t(a,b) \mid \mathcal{H}_t] - \hat{\mu}_{ab,t})$ and the $\hat{\mu}$ terms cancel, so $\mathbb{E}[\phi_t \mid \mathcal{H}_t] = \theta_t(X_t)$ for any $\mathcal{F}_{t-1}$-measurable $\hat{\mu}$. One indicator is nonzero, contributing at most $1/\pi_{\min}$, and $|\sum c\,\hat{\mu}| \le 2$, so $|\phi_t| \le 1/\pi_{\min} + 2$.

A.2 Proofs for Part B

Why the clipped score fails (Example 1).

Let s_t be i.i.d. with the law in the example and bet a fixed $\lambda = 0.1$ on $u_t = \mathrm{clip}(s_t, -5, 5)$. Then $t^{-1}\log E_t \to 0.1\log(1-5\lambda) + 0.9\log(1+1.5\lambda) = 0.0565 > 0$ a.s., so $E_t \to \infty$ although $\mathbb{E}s_t = -0.65$. The plug-in bet behaves identically: $\hat{\mu}/\hat{v} \to 0.188$ exceeds the cap 0.1, so $\lambda_t = 0.1$ eventually. A data-dependent scale does not repair this: $\mathbb{P}(s = -100) = 0.01$, $\mathbb{P}(s = 1) = 0.99$ has $\mathbb{E}s = -0.01$, root mean square 10.05 and, with $b = 5$, $\mathbb{E}\,\mathrm{clip}(s/\mathrm{RMS}, -5, 5) = +0.0485$; and $\mathbb{P}(s = -1000) = 0.001$, $\mathbb{P}(s = 0.99) = 0.999$ with $b = 10$ has $\mathbb{E}s = -0.011$, RMS 31.6 and clipped mean $+0.0213$. Two sufficient conditions do exist. Writing $m_t = \mathbb{E}[s_t \mid \mathcal{F}_{t-1}]$, σ for the scale and $U_t = \mathbb{E}[(s_t - b\sigma)_+ \mid \mathcal{F}_{t-1}]$, $L_t = \mathbb{E}[(-s_t - b\sigma)_+ \mid \mathcal{F}_{t-1}]$, one has the exact identity $\sigma\,\mathbb{E}[\mathrm{clip}(s_t/\sigma, -b, b) \mid \mathcal{F}_{t-1}] = m_t - U_t + L_t$, so the raw null implies the clipped null if and only if $L_t \le U_t - m_t$. In particular, if the conditional law of s_t is symmetric about m_t, then $m_t \le 0$ implies the clipped mean is ≤ 0 for every predictable $\sigma > 0$ and $b > 0$ (the map $x \mapsto \int \mathrm{clip}(x+z)Q(dz)$ is nondecreasing and vanishes at 0 for symmetric Q). The null stated on the clipped score itself is always valid (supermartingale by the argument below), but it is not the scientific null.

Lemma 13 (Predictable bound) *For $\ell_c(e) = \min(e^2, c^2)$ and any y, p^1, p^2, $|\ell_c(y-p^1) - \ell_c(y-p^2)| \le c^2 - (c - |p^1 - p^2|)_+^2 \le \min\{2c|p^1 - p^2|, c^2\}$, and the first bound is attained.*

Proof $\ell_c(e) = h(|e|)$ with $h(r) = \min(r^2, c^2)$, which is $2c$-Lipschitz, and $||e| - |e'|| \le |e - e'|$; this gives the $2c|p^1 - p^2|$ bound. For the sharp bound take $r_1 = |y - p^1| \le r_2 = |y - p^2| \le r_1 + d$ with $d = |p^1 - p^2|$; the difference $\min(r_2, c)^2 - \min(r_1, c)^2$ is at most $\min(r_1 + d, c)^2 - \min(r_1, c)^2$, which is maximised over r_1 at $c^2 - (c-d)_+^2$. □

Proof of Theorem 8 and Corollary 9 By Lemma 13, $|s_t| \le b_t$ for every outcome, so $|u_t| \le 1$ and s_t is integrable. Because b_t and $\mathbb{1}\{b_t > \varepsilon\}$ are predictable (functions of predictions made before Y_t is seen), $\mathbb{E}[u_t \mid \mathcal{F}_{t-1}] = \mathbb{1}\{b_t > \varepsilon\}\mathbb{E}[s_t \mid \mathcal{F}_{t-1}]/b_t \le 0$ under H_0^{preq}. With $\lambda_t \in [0, \gamma]$ predictable, $1 + \lambda_t u_t \ge 1 - \gamma > 0$ and $\mathbb{E}[E_t \mid \mathcal{F}_{t-1}] = E_{t-1}(1 + \lambda_t \mathbb{E}[u_t \mid \mathcal{F}_{t-1}]) \le E_{t-1}$. Ville's inequality applies. For the capped process, $x \mapsto \min(x, C)$ is concave, so by Jensen $\mathbb{E}[E'_t \mid \mathcal{F}_{t-1}] \le \min\{E'_{t-1}(1 + \lambda_t \mathbb{E}[u_t \mid \mathcal{F}_{t-1}]), C\} \le E'_{t-1}$. For contemporaneous features, replace $\mathcal{F}_{t-1}$ by $\mathcal{G}_t$ in the conditional mean of u_t and use the tower property: $\mathbb{E}[E_t \mid \mathcal{F}_{t-1}] = \mathbb{E}[E_{t-1}(1 + \lambda_t \mathbb{E}[u_t \mid \mathcal{G}_t]) \mid \mathcal{F}_{t-1}] \le E_{t-1}$. The group case only changes the feature maps. □

Proof of Proposition 10 Let Π_S denote orthogonal projection onto the span of feature set S, $r_{\mathrm{add}} = Y - \Pi_{\mathrm{add}} Y$ and $\tilde{X} = X - \Pi_{\mathrm{add}} X$ for the cross-product block X. (a) The joint span is the orthogonal sum of the additive span and $\mathrm{span}(\tilde{X})$, so by Pythagoras $\mathrm{Syn}_{\mathrm{aj}} = \|\Pi_{\tilde{X}} r_{\mathrm{add}}\|^2 = \kappa^\top \Sigma_{\tilde{X}}^+ \kappa \ge 0$ with $\kappa = \mathbb{E}[X r_{\mathrm{add}}]$, which is zero if and only if $\kappa = 0$. (b) The identity $\mathrm{Syn}_4 = \mathrm{Syn}_{\mathrm{aj}} - R$ with $R = [L(\mathrm{base}) - L(A)] - [L(B) - L(\mathrm{add})]$ is algebra; if the centred blocks $\bar{F}_A, \bar{F}_B$ are orthogonal then $\Pi_{\mathrm{add}} = \Pi_W + \Pi_{\bar{F}_A} + \Pi_{\bar{F}_B}$ and $R = 0$. In the Gaussian example with $Y = A \pm B + \varepsilon$ and $\mathrm{corr}(A, B) = \rho$: $L(\mathrm{add}) = L(\mathrm{joint}) = \sigma^2$, $L(A) = L(B) = 1 - \rho^2 + \sigma^2$ and $L(\mathrm{base}) = 2 \pm 2\rho + \sigma^2$, which gives $\mathrm{Syn}_4 = -2\rho(1+\rho)$ for $+$ and $+2\rho(1-\rho)$ for $-$. (c) (S1): $Y - \varepsilon \in \mathrm{span}(\mathrm{add})$, so $r_{\mathrm{add}} = \varepsilon$ and $\mathbb{E}[X\varepsilon] = 0$. (S2): by independence,

centred own-block spans are mutually orthogonal, so $r_{\text{add}} = r_m(W) + r_f(\mathcal{A}) + r_h(\mathcal{B}) + \varepsilon$ with each term centred and orthogonal to its own block, and every product $\mathbb{E}[A_\ell B_\ell r_\bullet]$ factorises to zero. (S3): after standardisation expand $f = \sum a_k He_k(A)$, $h = \sum b_k He_k(B)$ in Hermite polynomials; by Mehler's identity $He_k(A)$ ($k \geq 3$) is orthogonal to the additive span, so $r_{\text{add}} = \sum_{k\geq 3}(a_k He_k(A) + b_k He_k(B)) + \varepsilon$; and $\mathbb{E}[AB\, He_k(A)] = \rho\, \mathbb{E}[A^2 He_k(A)] = 0$ for $k \geq 3$ since $A^2 = He_2 + He_0$. (d) For $B = A^2 + \tau V$, $Y = A^3 + \varepsilon$: the features $1, A^2, B, B^2$ are even in A and A, $X = AB$ odd, so $r_{\text{add}} = He_3(A) + \varepsilon$ and $\tilde{X} = He_3(A) + \tau AV$, giving $\kappa = 6$, $\mathbb{E}\tilde{X}^2 = 6 + \tau^2$ and $\text{Syn}_{\text{aj}} = 36/(6 + \tau^2)$. For the Gaussian two-lag example, $X = A_{t-1}B_{t-1} = A_{t-1}A_{t-2} + vA_{t-1}V_{t-1}$ is orthogonal to the additive span with $\mathbb{E}[XY] = 1$ and $\mathbb{E}X^2 = 1 + v^2$. □

Proof of Proposition 11 Since $|s_t| \leq b_t$ and $\delta > 0$, $w_t \in [(\delta - b_t)/(\delta + b_t), 1] \subset (-1, 1]$, so every factor is at least $1 - \gamma$, and $\mathbb{E}[w_t \mid \mathcal{F}_{t-1}] = (\delta - \mathbb{E}[s_t \mid \mathcal{F}_{t-1}])/(b_t + \delta) \leq 0$ under H_δ because $b_t + \delta > 0$ is predictable. Hence $K(\delta)$ is a supermartingale and Ville's inequality applies; a union bound over the grid gives the second statement. □

Proof of Proposition 12 Because $s_s \in [-b_s, b_s]$ with b_s predictable, Hoeffding's lemma applied conditionally gives $\mathbb{E}[\exp(\lambda(m_s - s_s)) \mid \mathcal{F}_{s-1}] \leq \exp(\lambda^2 b_s^2/2)$. Hence $Z_t(\lambda) = \exp\big(\lambda \sum_{s=t_1}^{t}(m_s - s_s) - \frac{\lambda^2}{2} V_t\big)$ is a nonnegative supermartingale with $Z_{t_1-1} = 1$, and Ville's inequality gives $\mathbb{P}(\exists t : Z_t(\lambda_j) \geq K/\alpha) \leq \alpha/K$. A union bound over j and rearranging $Z_t(\lambda_j) < K/\alpha$ gives the stated bound; the lower bound follows from $-s_s$. □

Estimation noise and Clark–West.

With squared loss, $\mathbb{E}[s_t \mid \mathcal{F}_{t-1}] = (\mu_t - a_t)^2 - (\mu_t - j_t)^2$ where a_t, j_t are the two forecasts and $\mu_t = \mathbb{E}[Y_t \mid \mathcal{F}_{t-1}]$. The Clark–West adjusted score $s_t + (a_t - j_t)^2$ has conditional mean $2(\mu_t - a_t)(j_t - a_t)$, which is zero only when $a_t = \mu_t$ and can otherwise have either sign; it is therefore a diagnostic and not a valid centred score.

Appendix B Average-synergy bounds by family

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Table B1 Time-uniform upper bound of Proposition 12 on the running-average conditional synergy of the learned forecasters (loss units per step), by family of the financial search (daily and weekly panels). m: hypotheses; "mean s": median over hypotheses of the realised mean score; the bound uses level $0.05/m$ and $K = 22$ grid points; "median" and "max" are over the hypotheses of the family

Panel	Target	Structure	m	median mean s	median bound	max bound
etf16	ret	lag1	1365	−0.006	0.087	0.77
etf16	ret	lag5	1365	−0.027	0.391	11.85
etf16	ret	contemp	1365	−0.009	0.097	0.61
etf16	ret	weekly	1365	−0.014	0.287	1.08
etf16	vol	lag1	1365	−0.006	0.079	1.50
etf16	vol	lag5	1365	−0.026	0.361	8.31
etf16	vol	contemp	1365	−0.008	0.104	1.69
etf16	vol	weekly	1365	−0.014	0.257	1.91
stocks13	ret	lag1	660	−0.006	0.070	0.62
stocks13	ret	lag5	660	−0.030	0.417	17.39
stocks13	ret	contemp	660	−0.011	0.092	0.48
stocks13	ret	weekly	660	−0.016	0.231	0.77
stocks13	vol	lag1	660	−0.006	0.065	1.76
stocks13	vol	lag5	660	−0.028	0.433	19.61
stocks13	vol	contemp	660	−0.008	0.122	2.15
stocks13	vol	weekly	660	−0.015	0.215	0.98
world6	ret	lag1	30	−0.005	0.065	0.15
world6	ret	lag5	30	−0.034	0.751	8.37
world6	ret	contemp	30	−0.009	0.074	0.17
world6	vol	lag1	30	−0.005	0.057	0.11
world6	vol	lag5	30	−0.024	0.617	6.37
world6	vol	contemp	30	−0.008	0.081	0.26

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